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Estimating dynamic population using cellular network data

Master Thesis

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Abstract

Obtaining precise and up-to-date population data holds immense potential for multiple fields such as public health, urban planning, disaster management, and understanding societal changes. Traditional census methods used to estimate these numbers often fall short, becoming quickly outdated, necessitating the exploration of alternative approaches for dynamic population estimation. Mobile network data is one data set that seems well-suited for these new approaches, given the ubiquity of mobile phones and their tendency to always be carried with the person, along with their ability to provide precise spatial location data, making them a valuable resource for assessing population distributions. To tackle this challenge, this paper seeks to address the question of whether methodologies from the field of geoinformatics can be used to estimate both static and dynamic populations using mobile network performance data. Supplementary questions are the possibility of using an existing method with higher spatial resolutions, which kind of data is the most suited for the estimation and if the study area can be increased in size. The used methodology utilized privacy-compliant mobile network data together with CORINE land cover and building coverage data to built two models that are able to predict the residential night population and the dynamic population at any point in time on a spatial scale of 100m by 100m using mobile network data. The collected data includes the four metrics active users, voice calls, Erlang and data usage. The study encompasses the city of Munich and its surrounding regions. Key findings are that both models are able to predict the spatial distribution of the population, but the population is underestimated, especially for the dynamic prediction. Additional results are that while a general model is able to predict a large area, models trained on smaller subsets can be used to estimate populations in different areas, with higher accuracy than the general model. A further important insight is that the initial distribution of data is of a high importance for model quality. Furthermore, the use of raw performance data, rather than more extensively pre-processed data, is also confirmed as a viable approach, as is the utilization of the predefined CORINE separation layer. Another important aspect is that verifying the results at a small spatial scale becomes increasingly difficult because there is limited and outdated ground truth data available. In summary, this study can confidently state that methods from the field of geoinformatics can estimate dynamic and static populations using mobile network data.

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1 Introduction

Population estimation has long been an important endeavor for understanding societal dynamics and resource allocation. Post-modernity brought a compaction of space-time, lowering travel cost significantly and making communication costs time-invariant. Harvey (2000) states that this compaction leads to a more flexible society, with higher fluctuations of individual mobility. Acknowledging this highly complex and dynamic system and trying to understand and estimate the spatial distribution of the population would allow for more effective policies in almost all departments of governance, be it health, food security, emergency response, education, natural disasters, infrastructure and transportation.

Krisp (2010) discusses that knowing the present population would enable fire departments to estimate actual populations apart from rigid census data and administrative boundaries and allow for more effective planning. From an economic standpoint, this data could yield valuable insights, benefiting tasks like location analysis or the estimation of foot traffic. A practical example case would be an analysis for changes in population mobility due to the COVID-19 pandemic. The German Federal Statistical Office was able to detect a significant reduction in mobility in the years of the pandemic and evaluate the effectiveness of lock-down measures using mobile network data.¹ Willberg et al. (2021) detected a similar effect due to COVID-19 in Finland using comparable data network data.

However, the existing census data is often already outdated after a short time because of people relocating or newly constructed buildings and is only updated in large intervals. It also usually does not represent the actual population, but only the residential population. Also, according to Douglass et al. (2015), a census is expensive, has a large administrative overhead, and sometimes is politically resisted by the population. They recommend finding alternative methods.

However, these methods for estimating the population spatially and temporally are increasingly challenged by the growing dynamism and flexibility of society. The advent of mobile technology after the year 2000 and the growing availability of mobile devices allows for novel opportunities to address this challenge. Because these devices are often carried and used throughout the day, they serve as a powerful means to comprehend the complexities of modern society. Daily routines, social interactions and travel patterns can be discerned using this data. They serve as a reflective mirror of society, capturing diverse aspects of human behavior. Analyzing this data can lead to a better understanding of collective and individual behaviors and trends.

¹<https://www.destatis.de/DE/Methoden/WISTA-Wirtschaft-und-Statistik/2021/03/mobilitaet-corona-pandemie-032021.html>, Accessed: 05.03.23023

Due to the high penetration rate of mobile devices, almost every human can be seen as a data point in a highly decentralized, global sensor network. There are over 177 million active SIM-cards in Germany in 2022, divided into roughly thirds between the three big network providers Vodafone GmbH, Telekom Deutschland GmbH and Telefónica Germany GmbH & Co. OHG (ordered by number of active SIM-cards).² As the number of network activity is steadily rising, with more calls and especially a higher mobile data usage, the importance and accuracy of this data will also rise.

One of the useful potential insights that could be derived from this data is the dynamic population at any specific place at any specific time. However, estimating the actual population from mobile network data remains an ongoing effort within the research. The research community currently lacks consensus regarding the optimal methodology and many open questions remain. These include the feasibility of developing a nationwide or even global model and the possible spatial resolutions. Possible applications of such a model are diverse. One practical use case involves the implementation of energy-saving measures within the telecommunications industry. Since the end of 2022, mobile network providers are allowed to temporarily deactivate network cells in low-traffic areas to save energy.³ A dynamic population model could easily predict the times where a region is devoid of any population, thereby enabling network cell shutdowns without compromising service quality.

In light of this context, this paper focuses on the central research question: Can the dynamic population be estimated from mobile network data using methods from the field of Geoinformatics? An additional spotlight is placed on the question of how well the existing method can be applied to higher spatial resolutions and a larger study area with a larger share of rural areas. It will also be discussed which kind of network usage data is most suited for population estimation. Furthermore, this paper uses raw data without personal identifiers which most other similar studies do not. This renders the data privacy-compliant, which is important, because non-personal data is easier to use and acquire, and therefore streamlines following research.

In order to address these questions, a case study is carried out which uses raw mobile network performance data and a pre-established methodology from Khodabandelou et al. (2019b) for estimating dynamic populations. The results of this case study are then examined with regard to the performance and suitability of the used methodology.

²<https://www.bundesnetzagentur.de/DE/Fachthemen/Telekommunikation/Marktdaten/Mobilfunkteilnehmer/artikel.html>, Accessed: 21.02.2023

³Source:<https://www.zdf.de/nachrichten/digitales/bundesnetzagentur-mobilfunk-energiesparmodus-100.html>, Accessed: 02.02.2023

The used methodology first spatially distributes the collected mobile network performance data. It then builds an univariate linear regression model to estimate the static night population. It then builds on top of this a second, time-variant, multivariate dynamic population model.

This paper first outlines the theoretical foundations of the used technologies and central concepts like night and dynamic population. Then a comprehensive overview of the current state of research for estimating dynamic populations with cellular network data follows. The subsequent case study chapter firstly explains in the data chapter the used sources and goes into further detail examining the uncertainties associated with the data. The following methodology chapter outlines the used approaches for estimating the night and dynamic population. This leads to the results section, which presents the outcomes of the two models, examines them using different evaluation metrics and presents examples. The subsequent discussion examines the results and addresses limitations and shortcomings. Finally, the conclusion and outlook chapter reflects the implications from the case study, and places them into the wider context of the research question. It then expands on gained insights and outlines potential avenues for subsequent inquiry.

2 Theoretical Foundations for using Mobile Network Data to predict Dynamic Populations

This chapter provides the theoretical foundation to understand the technologies and concepts referred to in this thesis. Firstly, the technological aspects of cellular networks are explained. Secondly, a chapter on modern usage of mobile networks follows. After that, the concepts of night and dynamic population are defined. At last follows a short overview over analyzing mobile phone data with regression.

2.1 Mobile Network Technology

Mobile network technology can be divided into three parts. The mobile station, the radio access network (RAN) and the core network or backbone. The interaction of these parts can be seen in figure 1a. The mobile station is the device used to communicate over the mobile network. These can be mobile phones, internet of things devices, cars, or any other type of device with a broadband adapter. They are known as user equipment, or UE (Zhao, 2002). The RAN is the part of the network that communicates between the mobile station and the core network. The different RAN sites are connected to the core network using the backhaul.

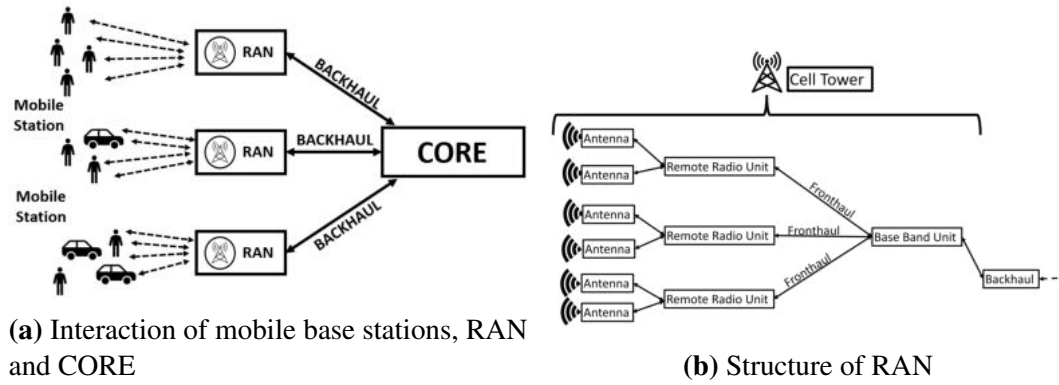


Figure 1: Mobile network architecture

This is usually achieved with high-speed glass fiber or microwave radio antennas (Sauter, 2022: p.176). In addition, the Core network connects all RAN-Sites with external networks like the internet (Sauter, 2022: p. 176).

The RAN contains the most visible parts of a mobile network, being the mobile network sites, commonly known as cell phone towers (Sauter, 2022:p. 120). They are also called node B for 3G and eNode B for 4G and 5G networks (Sauter, 2022:p. 6). One cell phone tower can have antennas for different frequencies and technologies that transmit radio signals into different directions. Each combination of antenna orientation, frequency and technology is called a network cell. A single site can have multiple antennas for each frequency sending to different directions. As can be seen in figure 1b the first element of a site is the baseband unit (BBU), which manages the different network cells and decodes and encodes the digital signals (Sauter, 2022: p. 120). This unit is connected on one side to the backhaul, the other side connects to the fronthaul, which transmit the digital signal to the remote radio units (RRU), typically using a fibre connection (Sauter, 2022: p. 125). These either convert the digital signal from the BBU to an analog signal and send it to the antenna or they convert the analog signal from the antenna to digital. The antenna unit, consisting of multiple smaller antennas, is connected to the RRU and transmits and receives the radio signals (Sauter, 2022: p. 120).

The RAN is the user-facing side of a mobile network and thus in control of maintaining all aspects of user connections. This means load balancing, managing the frequency spectrum and general resource allocation. Increasing the RAN performance leads to the most significant improvements in a mobile network (Zanferrari, Morais, Fernando et al., 2020). The radio connection from the node B to the user equipment is called downlink, whereas the other way is called uplink (Zhao, 2002). Any active mobile device is connected to a number of different cells at the same

time, and the data can send using different cells and frequency bands concurrently, a process called carrier band aggregation (Sauter, 2022: p. 66 f.). If a mobile device moves from the area of one cell to another, the connection is also switched between these cells, a process called handover (Sauter, 2022: p. 10).

Each cell sends data on one specific frequency band between 700 MHz and 3800 MHz. Bands between 700 MHz and 900 MHz are in the low-band-spectrum, 1800 to 2100 MHz are mid-band and the 2600 MHz and above frequencies are called high-band. In addition, 5G networks send in the 3500 MHz range (Sauter, 2022: p.116 f.), and 26000 MHz bands are planned⁴. The bands are technology-agnostic, and can be used for for all three digital standards 3G, 4G and 5G concurrently (Sauter, 2022: p. 118). Different frequencies have different characteristics, with lower bands usually allowing for higher range but smaller bandwidth, which makes them more suited for use in areas with a lower, more spread out population, and higher bands having a short range but higher bandwidth, which makes them more suited for use in cities. These spectral bands are divided into small sections between 10 and 100 MHz, and each of these is reserved for one of the three major telecommunications providers: Telekom, Vodafone, and Telefónica (Sauter, 2022: p. 116 f.). A newer entrant in the market, 1&1 AG, has also secured its own spectrum to use for it's 5G network. All these providers hold portions of the radio spectrum across several frequency bands.

One challenge in this paper is to correctly assign the measured data spatially. This mobile positioning can be broadly divided into two categories. The first is using the network site locations as the spatial basis, by using Cell identification (Cell ID) and triangulating the mobile devices location by using the distance and angle to nearby sites. The other option is using device-based methods like the Global Positioning System (GPS) (Ahas, Mark, 2005). There are techniques for increasing the spatial accuracy of the Cell ID method, like using the travel time of the radio signal with the Timing Advance (TA) method or the angle of the arriving signal with the Angle of Arrival (AOA) method (Spinney, 2003). More advanced methods like the Enhanced observed Time Difference (E-OTD) use data from multiple connected base stations and use complex propagation time calculations for a more accurate estimation (Mateos, Fisher, 2007). There are many more methods, like Time difference of arrival (TDOA) or Observed time difference of arrival (OTDOA) (Zhao, 2002), but they all calculate the position of a specific mobile device. These methods are thus not privacy-compliant as required in this paper.

⁴<https://www.bmuv.de/faq/welche-frequenzen-nutzen-die-5g-netze>, Accessed: 28.04.2023

2.2 Antenna Technology and Coverage

Mobile devices try to connect with the nearest network cell, because signal power decreases with distance. Each cell is surrounded by an area where its antennas signal is stronger than that of other cells. Often this is simplified and aggregated to the network site level. Then the coverage area from all cells of one network site are aggregated to one uniformly sized hexagon, with a omnidirectional antenna in the middle (Miao, 2016: p. 95). Figure 2 shows a simplified comparison of five different ways of assessing a cellular network division. Figure 2a shows the division into hexagons, centered on each network tower. Modern cellular networks use directional antennas, where the hexagons are either divided into multiple cells or there is a hexagon for each cell, as shown in figure 2b, where each network tower has three network cells, servicing one hexagon each (Sauter, 2022: p. 419 f.).

Real world scenarios include sites with any number of antennas, with varying beam widths and different frequency bands. Also, not all sites are towers, but they can also be on high roofs, walls, tunnels or inside of buildings. Moreover, the actual signal propagation is also influenced by factors like diffraction, scattering, reflection or refraction. Diffraction describes the bending of the signal around an object which is bigger than the signals wavelength. Scattering on the one hand happens if the signal encounters objects smaller than its wavelength and spreads the signal out from its intended direction. Reflection on the other hand changes the signals direction. Finally, refraction happens due to the fluctuating atmospheric refractive index, which changes the signal strength (Erunkulu et al., 2020).

This leads to the popular approach of spatially locating cell phone events with Voronoi polygons. In this approach each event is assigned to the mobile tower nearest to its location. According to Ricciato et al. (2020), the most common Voronoi method is centering the polygons around the tower, like in figure 2c. Another variant of Voronoi tessellation is shown in figure 2d. Here, as described in Ricciato et al. (2020), the position of each cell's antenna is offset a few meters in the direction of emission, enabling a division into individual cells and allowing for a more granular analysis. A similar approach would be calculating the best-server polygons as described in the next paragraph, and then taking theses polygons centroid as a seed for the Voronoi tessellation.

However, this approach ignores that different cells can have different transmit power and therefore ranges can overlap. To be specific, one network site can have multiple antennas in the same direction with different ranges. Therefore, a mobile device does not necessarily connect to the nearest cell, but to the one with the best service quality (Ricciato et al., 2020). Because of this, the exact dimensions of a

network cell can only be estimated using complex propagation models (Erunkulu et al., 2020). These models create maps called Best Server Area (BSA). In these maps each point is assigned to the cell with the strongest signal. This also allows for the depiction of non-continuous coverage areas. An example for this kind of map can be seen in figure 2e. Here each raster cell is colored depending on which cell has the strongest signal. A variation of BSA maps are N-BSA maps. Meaning, each cell is assigned to the N strongest cell, where N can be any number (Ricciato et al., 2020).

2.3 Overview of the different Mobile Network Standards

This section describes the differences between the various standards of mobile network technology. The development of this technology is divided into five generations, 1G to 5G. In most countries, any combination of the standards 2G, 3G, 4G and 5G are currently in use and often antennas for all technologies can be found on the same site on different bands (Sauter, 2022;p. 73 ff.).

Figure 3 shows the number of base stations of each mobile network generation in Germany between the years 2019 and 2021. The data is from the Jahresbericht Telekommunikation from the German Bundesnetzagentur.⁵ In 2021, there is a marked reduction in the number of 3G antennas, falling almost to zero, owing to the deactivation of the 3G network. Having said that, the number of 4G and 2G stations stayed roughly the same since 2019, whereas the segment of 5G stations is growing. This data shows that structure and capabilities of the mobile network are continuously changing, and any study using data from this network needs to address this constant mutability.

⁵https://www.bundesnetzagentur.de/DE/Allgemeines/Presse/Mediathek_3/form.html, Accessed: 04.02.2023

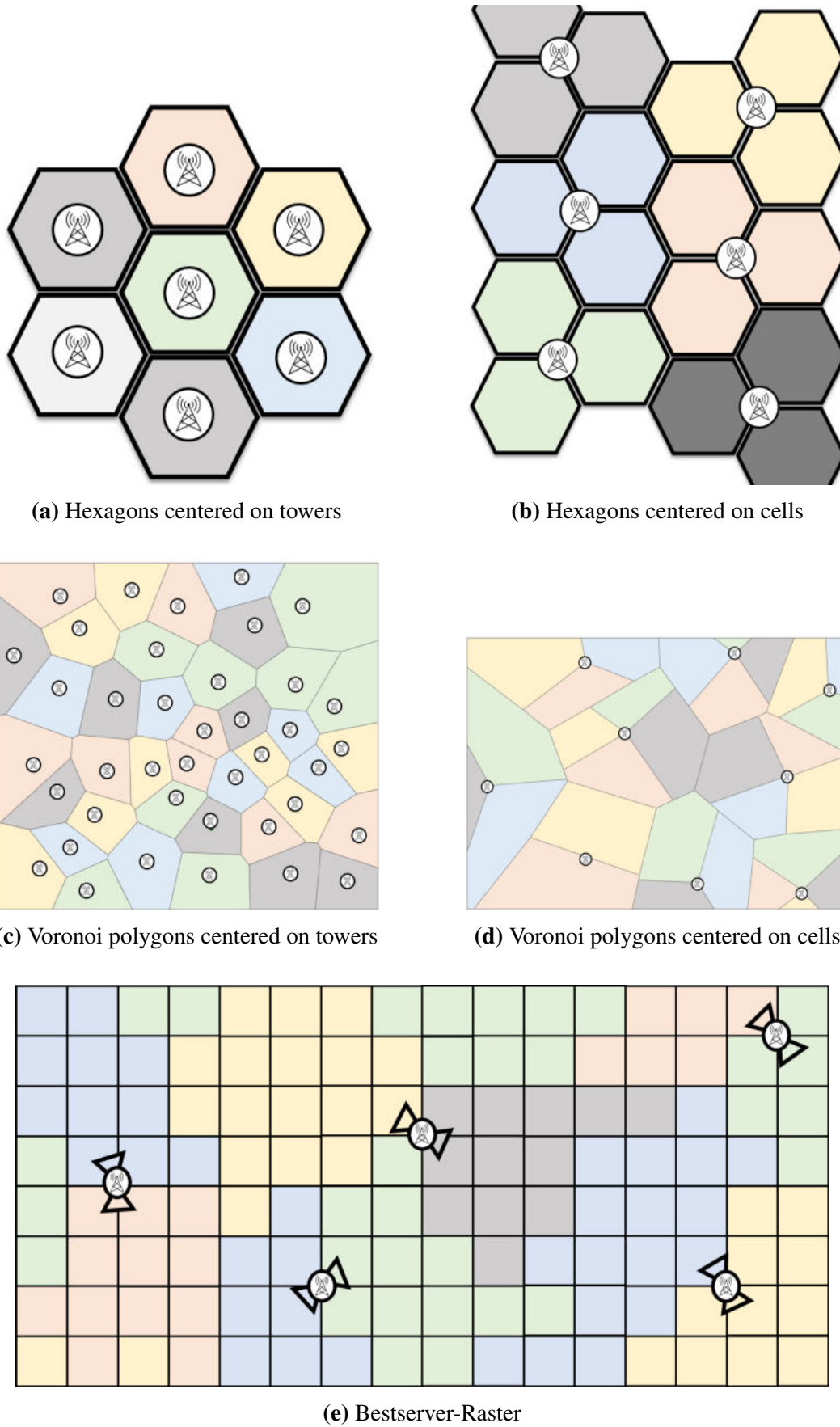


Figure 2: Different ways of visualizing network cells with varying complexity

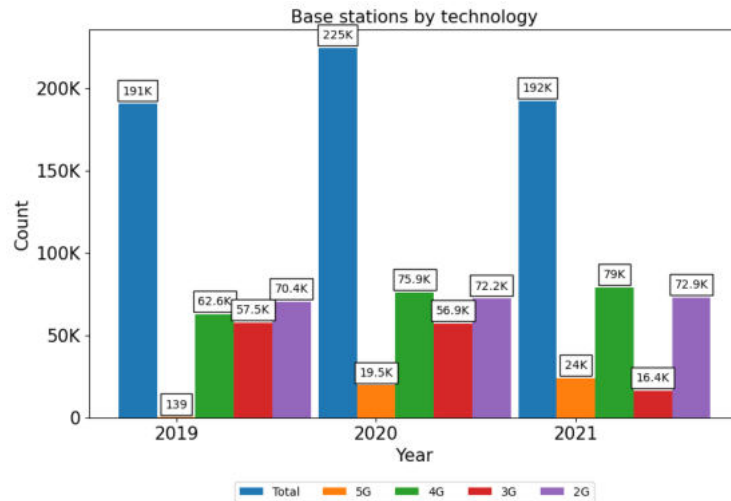


Figure 3: Base station count of the different mobile network standards active in Germany

The first generation of cellular networks, sometimes called 1G, was purely analog, and used similar routing to landlines. In Germany, there were three generations of analog or 1G networks: The A-, B- and C-Networks. The A and B networks were used between 1959 and 1977 and 1972 to 1994⁶ respectively. These networks were no true mobile networks, because they did not support handover. This means that the phone could not be moved to another network cell during a call (Kreis, 1989:p. 54). This was solved with the C-Net (1985-2000¹), which was thus the first true mobile network in Germany, because it allowed for unimpeded movement without service interruptions. Moreover, it also allowed for roaming, which means transferring calls between different networks and countries (Kreis, 1989: p. 56 f.). Subsequently, new network standards were developed.

2G - Global System for Mobile Communications (GSM)

The first of these new standards were the second generation mobile network systems, called 2G. The most notable ones are GSM in Europe, IS-95 in the USA and PDC in Japan (García et al., 2020). GSM, or Global System for Mobile Telecommunications was used between 1987 and 1989. It was the first non-proprietary digital standard after 1G, and allowed interoperability between different network providers. With the need for mobile data, 2.5G technologies were developed that supported wireless internet traffic. In Europe, the successor to GSM was General packet radio service (GPRS) (Dunnewijk, Hultén, 2007). In Germany GSM networks are still

⁶<https://www.informationszentrum-mobilfunk.de/technik/geschichte-des-mobilfunks/1-generation-1g>, Accessed 24.03.2023

maintained to ensure network access for old devices which are incompatible with newer standards and for emergency calls (Sauter, 2022: p. 73).

3G - Universal Mobile Telecommunications System (UMTS)

With a further need for higher speeds, more bandwidth and more concurrent users, a new standard was needed. The newly established Third Generation Partnership Project (3GPP) defined the Universal Mobile Telecommunications System (UMTS) standard in the year 2000. This technology allowed for packet switching and low, medium and high data rate connections (García et al., 2020). With UMTS, providers also started using multiple radio bands at the same site, increasing the complexity of the hardware and the available bandwidth (Sauter, 2022: p. 72). Moreover, UMTS was the first standard allowing for high data-rates, and thus enabling many new mobile applications.

4G - Long-Term Evolution (LTE)

Smartphones with high processing capabilities lead to an increase in high data volume applications. The 4G standard, defined in 2004 called Long Term Evolution (LTE), unified the voice and data channels, and could reach data rates of up to 1 Gbps (Condoluci, Mahmoodi, 2018). Because of the higher efficiency, many 3G channels were repurposed for 4G. As a result, 3G networks were shut down in Germany in 2021 (García et al., 2020). Thus, 4G networks are currently the backbone of most mobile networks. In addition, LTE is also the first standard that is the same in all countries, and thus allows easier roaming and use of unified hardware components (Sauter, 2022: p. 1 ff.).

5G - New Radio (NR)

Many modern applications have a need for extremely low latency, and the ability for the network to connect to a high amount of devices at the same time. There is also a growing segment of non mobile-phone devices using cellular networks. Examples are Internet of things (IoT) devices and internet enabled cars (García et al., 2020). For these cases, 5G enables the division of frequency bands into distinct segments governed by specific rules. This approach allows for the allocation of one segment to high-data-rate and high-latency traffic, while simultaneously dedicating another segment to low data-rate, low-latency traffic (Sauter, 2022:p.115). In order to introduce 5G faster into the networks, a hybrid solution called 5G non-standalone (NSA) can be used. In this case the 4G network is still used to connect the mobile device to the network, but the traffic is then passed along to the 5G network (Sauter, 2022: p. 116).

In addition, the 5G standard also includes features such as multiple input, multiple outcome (MIMO), beamforming, cloud computing, network virtualization and a good ability to scale with increasing demand (García et al., 2020).

Furthermore, 5G technology can use multiple paths at once, by scheduling data traffic in specific time-steps for different paths. This multipath propagation is a wireless network characteristic where the connection between mobile device and network cell can be established with multiple, different paths. One path could be a direct line-of-sight path, whereas another, longer path could be achieved by reflecting of a nearby wall. (Speidel, Speidel, 2021: p. 67 ff.).

2.4 Mobile Phone Usage Patterns: An Overview

This chapter gives at first a short overview over the mobile network usage in Germany, and then compares the usage of Short Message Service (SMS) messages, calls and mobile data. All mobile devices can only be used with an Subscriber Identity Module (SIM) card, which uniquely identifies the device. It has to be considered that SIM-cards can usually be freely switched between different devices. For a better overview, figure 4 shows registered sim cards in Germany between the fourth quarter of 2001 and the fourth quarter of 2022 for the three cell network providers Telekom, Vodafone and Telefonica. The data from Telefonica before 2014 includes data from Telefonica and E-Plus, both companies merged in 2014. The data before 2014 is reported yearly, the data after 2014 quarterly. It must be noted that the data is self-reported by the different telecommunication companies to the Bundesnetzagentur⁷. Moreover, this data does include unused SIM-cards that are still registered by the providers. Furthermore, it also does not show the number of persons owning a SIM-card, as one person can own any number of cards.

As can be seen, the number rises steadily since 2002. Since 2006, the number of active SIM-cards in Germany is higher than the population. The three providers today have similar amounts of active cards, between 45 Million for Telefónica and 55 Million for Vodafone in 2022. The big jump of Vodafone SIM-cards in the second quarter of 2016 is due to them also considering Machine to Machine SIM-cards.

Not only did the amount of SIM-cards change, but also the general usage. Figure 5 shows a comparison between the usage of Short Message Service (SMS), phone calls and mobile data per active SIM-Card and per month in Germany between 2016

⁷<https://www.bundesnetzagentur.de/DE/Fachthemen/Telekommunikation/Marktdaten/Mobilfunkteilnehmer/artikel.html>, Accessed 30.03.2023

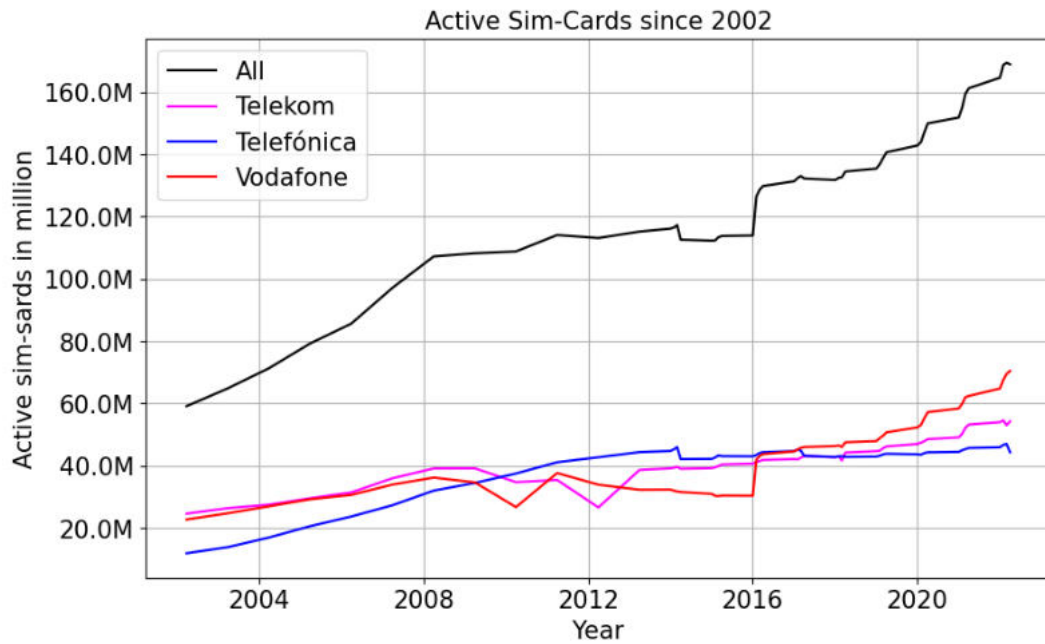


Figure 4: Active SIM-cards in Germany

and 2020. The numbers are collected from the "Tätigkeitsbericht Telekommunikation 2021" from the German Bundesnetzagentur⁸. As can be seen, voice calls (red) are steadily rising since 2017, with around 88 calls per SIM-card and month in 2017 and over 120 calls per card in 2020. In contrast to this, the amount of SMS is steadily declining, from ten SMS per card and month in 2016, to five in 2020. This change can be attributed to the rise of internet messaging applications, that allow sending short messages similar to SMS over the mobile network. Furthermore, data traffic is rising steadily similar to the voice calls, from less than 500 Megabyte (MB) per month and card in 2016, to over 3,000 MB in 2020. This increase can be attributed to the growing usage of data-heavy applications and more integrated media like high-resolution images and video. However, this heightened usage isn't evenly distributed among all users. Instead, a minority of users account for the majority of data consumption, while most users have relatively low mobile data usage (Jin et al., 2012).

In order to evaluate the different traffic volumes of calls, SMS and Data correctly, one has to consider that these metrics are usually measured differently. With SMS as a total number, Data traffic in gigabytes of Up- and Downloads and Voice traffic in Erlang. One Erlang equals one total hour of call traffic. But today, all traffic, be it data, voice or SMS is routed with the same digital network protocol. Thus for the provider, the total data rate, which includes data generated by user-data traffic,

⁸<https://www.bundesnetzagentur.de/SharedDocs/Mediathek/Taetigkeitsberichte/2021/Telekommunikation2020.pdf>, Accessed: 02.04.2023

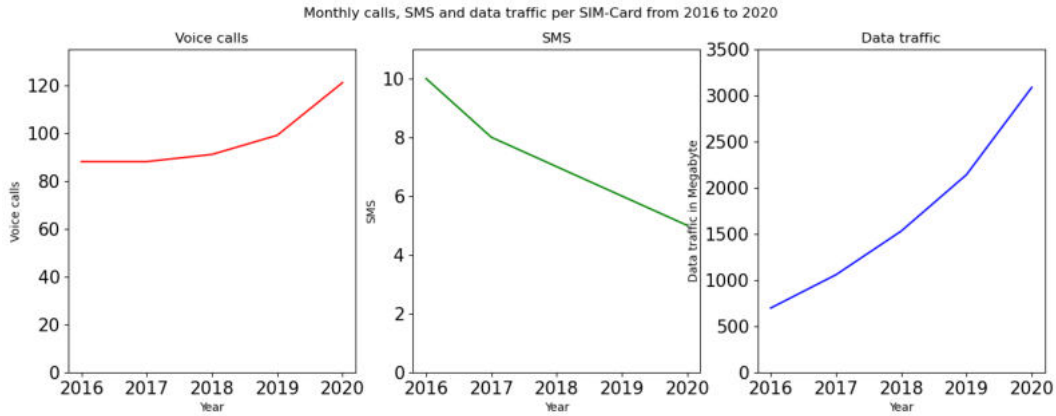


Figure 5: Voice, SMS and Data traffic per SIM-Card and Month in Germany

voice calls and SMS is the most important value for workload measurement. Generally these metrics also have a different daily seasonality. For example, Data traffic is much more homogenous, with Calls and SMS having a more heterogenous day pattern with more striking peaks (Soto, Frías-Martínez, 2011).

2.5 Estimating Population with Mobile Network Data

When looking at human populations, it has to be addressed that they are not static, and there are temporal and spatial fluctuations and trends. Post-modernity brought distance-invariant communication, enabling low-cost transmission of information over any distance. As mentioned earlier, travel costs also decreased, reducing the amount of barriers and strengthening a dense global urban system. These two factors lead to a compaction of space-time, further increasing the fluctuation of personal mobility (Harvey, 2000). Population dynamics also became more complex with the advent of mobile phones and mobile internet usage, because there are more activities that can be performed on the go or at a broader range of locations today (Ratti et al., 2006). Therefore, physical mobility today is intrinsically linked with and competes with virtual mobility. Thulin, Vilhelmson (2012) point out, that the decision between, for example, visiting a local store or online shopping leads to more flexible and dynamic mobility behaviours in space and time.

Another important factor to consider is that there are different kinds of populations, with different challenges to note when trying to estimate them. In this paper, the focus is on two kinds of population. The first one is the night population and the second one the dynamic population. The night populations can serve as a useful comparison to classical census data, whereas the dynamic model can provide in-

sights into transient population dynamics.

The night population in this thesis is defined as everyone with permanent residence at a specific location. This population is usually captured with a census. The most common census methods anchor populations to one place. As Hugo (2007) underlines, there are a lot of variations in the actual locations that a person resides in at any point in time. There are differences between day and night, between work-days and weekends, between holidays and non-holidays and seasonal fluctuations (Hugo, 2007). This variation is captured by the second relevant population in this paper, the dynamic or day time population which follows the definition in Ricciato, Coluccia (2023), as encompassing all individuals who are physically at a specific place at a specific time. Including commuters, tourists, students and other temporary visitors. This population can not easily be determined using classical census based methods and methods like manual counting or image-based recognition are usually only used in specific locations, and not in an extensive spatial and temporal manner. Furthermore, this population is more complex to estimate, because it includes large-scale, large-distance movement as well as more localized, daily travel patterns (Hannam et al., 2006).

There are numerous ways of estimating population using different kinds of data. These can be the use of nighttime satellite images such as in Sutton (1997) or Dobson et al. (2000), Road networks as in Xie (2006), vegetation coverage as in Pozzi, Small (2005) or the density of points of interest as in Doll et al. (2006) and Bakillah et al. (2014). One data set that is well suited to estimate dynamic populations is mobile network data. Ratti et al. coined the term "Mobile Landscapes" in 2006 to describe these new ways to understand urban systems and acknowledged the new possibilities to measure personal mobility on a level and precision impossible until then.

This mobility through the mobile landscapes can be tracked using mobile network data, and then be extrapolated because individuals often have highly regular travelling schedules. Also, most individuals have only a few locations that they return to regularly. These include their residence, place of work or education and recreational facilities (González et al., 2008). It can be said, that human travel behavior is highly regular, and this regularity is independent from the distance traveled. In addition, the majority of the population stays in a small area most of the time, and only a small minority travels greater distances. Interestingly, the regularity of human movement is also not imposed by any work schedules, but stays similarly regular on the weekends. Song et al. (2010) add that this predictability of movements does not significantly differ between different demographics.

This predictability makes the detection of spatial-temporal movements in the population easier. Migration as well as daily movement patterns can be detected (Kamenjuk et al., 2017). One well known daily pattern is that mobile phones are usually more used in the day time than at night and there are more calls on workdays than on weekends (Kang et al., 2012).

2.6 Analyzing Mobile Data with Regression

According to Bettencourt (2013), most properties of cities depend on population size. The correlation can vary, with some properties having a linear relationship, and others a super- or sublinear relationship with population (Krings et al., 2012). The relationship between population and mobile telecommunication activity can be described by a power-law relation like in equation 1 as used in Deville et al. (2014), Douglass et al. (2015) and Bettencourt (2013). In this equation, p is the estimated population, σ is the mobile phone activity and α and β are the parameters fitted by the model. By transforming the used data to a logarithmic scale, this relation can be represented quasi-linearly and modelled using regression.

$$p = \alpha \sigma^\beta \quad (1)$$

Regression analysis, introduced in 1885 by Galton, is a common method to model the effects of a set of independent variables on an other set of dependent variables. The relationship described between these variables is not deterministic, but has random errors and is only an estimation (Fahrmeir et al., 2015:p. 21). The models can also be used to determine the impact of each independent variable on the dependent ones. Moreover, regression models can differ in the type of variables, which can be continuous, binary, categorical or a count (Fahrmeir et al., 2015:p. 22). An example for a simple linear regression model is given in equation 2. This model establishes a relationship between the independent variable or regressor X and the dependent variable or regressand y . E , for error, represents all factors effecting y except X . C is the slope variable, because it represents the slope of the line that is the best fit of the model. It is also called the coefficient, because it multiplies X and thus scales the magnitude and direction of X . I is the intercept, also called constant term. It is the value the model predicts if X and E are zero (Wooldridge, 2009: p.23).

$$y = I + C * X + E \quad (2)$$

The standard method for fitting models using linear regression is the ordinary least squares method. This method fits the model by trying to minimize the residual sum

of squares (Wooldridge, 2009: p. 27 f.). This means the sum over all error terms E squared should be minimal. This regression technique is used in some of the papers introduced in the following chapter as well as in the methodology of this work.

3 State of Research in Mobile Network-Based Population Estimation

The previous section laid out the foundational concepts of mobile networks and using their data to estimate populations. This section will explore the current state of research for using mobile network data to predict population dynamics.

In 2014, Deville et al. published an influential paper giving a comprehensive overview over the possibility of mapping dynamic population using mobile phone data. They compare a mobile phone data based approach with a remote sensing method. They used Voronoi polygons to spatially locate calls and linear regression to fit their model. Moreover, they used nighttime data to estimate the night population, and day-time data to estimate daytime populations. They also divided the data into workdays and weekends. Their result was that their remote sensing method has a higher precision, but a lower accuracy than the mobile network data method. They conclude that mobile network data can be used to cost-effectively estimate population dynamics which supports the research question (Deville et al., 2014).

In 2012, the company TIM S.p.A. launched a big data challenge, releasing a large data set containing mobile network data from the Italian cities of Milan and Trento to the scientific community⁹. A large amount of papers in the field of estimating populations using this data set were released in the subsequent years. One of the first was by Douglass et al. in 2015. They use non-personalized metadata to predict static and dynamic populations on a base spatial unit of regular 235m by 235m cells. Their prediction algorithm is based on linear regression algorithm, similar to Deville et al. (2014), for the night-population and random forest regression for the dynamic prediction while using the metrics of incoming and outgoing SMS, voice calls and internet volume. They assess that the outgoing calls are the metric with the highest correlation to the underlying census population data. Douglass et al. (2015) summarize that the accuracy of the model is much higher if the population of the underlying spatial unit is above a certain threshold. Thus predicting the population using mobile network data is mostly possible in populous areas. This assertion underscores the challenge of extending the study area to encompass more rural regions.

⁹<https://theodi.fbk.eu/openbigdata/>, Accessed: 01.02.2023

Khodabandelou et al. (2019a) used the same data set from TIM S.p.A to predict the night and dynamic population. Khodabandelou et al. (2019b) later refined the approach. In contrast to Douglass et al. (2015), they used an irregular grid tessellation with grid cells between 255m by 325m and 2km by 2.5km. They compared the four different metrics voice, text, data traffic and subscriber presence measured in fifteen minute intervals. To fine-tune their model, they applied different layers of filtering and used the median week signature (MWS) method as described in Furno et al. (2015) and Furno et al. (2017) to split the study area into different land use classes. To predict the night population they use the linear RANSAC regressor with the same linear model as Deville et al. (2014). Their dynamic population model uses a new multivariate relationship between the subscriber presence and a subscriber activity metric. They come to the conclusion that their model can predict the typical daily trends and also has a good accuracy with large-scale events like soccer games (Khodabandelou et al., 2019b). Their paper is used as base for the methodology employed in this paper.

Building upon the historical context, the exploration now turns to a series of notable papers released in more recent years. Bachir et al. (2019) estimate origin-destination flows using mobile network call record data. The records have a temporal accuracy of one second, and are spatially located by cell-wise Voronoi polygons around network towers. They use a semi-supervised learning algorithm to infer the used mode of transport and then construct Origin-Destination matrices. Their model then estimates time-variant road and rail flows. They validate the results with ground truth travel survey and travel card data and conclude that their model performed well, with several issues, like the difficulty of separating private road use from public transport (Bachir et al., 2019). These findings indicate that approaches distinct from prevalent regression methods can also yield accurate outcomes.

Another study by Ricciato et al. (2020) formulated a framework for models estimating dynamic populations with mobile network data. They used Belgian data from a single day consisting of anonymized call data records, which allow tracing an individual user's mobility over the studied time span. The call data was spatially interpolated with four different methods: Bilayer Voronoi, Proximus Voronoi, Naif Voronoi and Overlapping locations, and used synthetic data to validate the results. They conclude that the geo-location method has a significant impact on the results, and in their case the overlapping locations approach had the most accurate results (Ricciato et al., 2020). This point was further emphasized in Ricciato, Coluccia (2023), where they compared their novel maximum a posteriori estimation method derived from a hierarchical generative model with more usual tessellation approaches. They used a semi-synthetic ground truth data set based on the city

of Munich and its surrounding districts to test their approach. In this study, their method showed better results than the Voronoi methods, while still maintaining an acceptable level of computational complexity. These studies underscore the role of pre-processing the data before the model and reveal notable accuracy differences among spatial distribution methods.

A slightly different approach is presented by Šveda et al. (2020) who used different data sets than most other studies in the field, being longitudinal signaling exchange (VLR) data. Their data, from a two week sampling period on the network cell level, contains a unique ID for each mobile phone, which means that movement of a specific phone can be traced. The data is spatially located by assigning it to the centroid of the approximated cell coverage or best server area. These centroids are then summed in a 1000m by 1000m grid raster. They use the anchor point model by Ahas et al. (2010) to filter out irregular visitors like tourists from the data. They conclude that VLR data is well suited to analyse dynamic population patterns but note that broad categories like night or daytime population describe individual spatio-temporal movement patterns only insufficiently. They advise to use additional data sources like traffic counters, footfall sensors and video tracking (Šveda et al., 2020). This could lead to the conclusion that a dynamic model, with the ability to estimate the population at any specific time, is needed to describe the actual population, instead of more general models.

The important matter of spatially locating the initial network data is addressed by Tennekes, Gootzen, Yvonne A. P. M. in 2021. They built a model that tries to estimate the location of mobile devices and compared different Bayesian methodologies. These methods are a uniform Bayesian distribution, a Land-Use weighted approach with four classes, a Network weighted approach and a composite method. They also incorporated timing advance data into their model. They concluded that a combination of using the composite data together with a signal-strength based network propagation model is most suited for future research (Tennekes, Gootzen, Yvonne A. P. M., 2021). This suggests that by integrating supplementary data alongside a more complex cell coverage algorithm, results could be improved.

Another aspect of spatial distribution is examined by Aasa et al. in 2021 using Estonian data. They used anonymized call data records to compare five spatial distribution methods. These are a simple point-in-polygon method, areal weighted interpolation, housing weighted interpolation, adaptive morton grid interpolation and random forest regression. They conclude that the point-in-polygon and areal weighted interpolation give poor results, because they only explain 42% and 35% of the variation in the data. The other three methods all have significantly higher determination coefficients with 80%, 95% and 93% for housing weighted, adaptive

morton grid and random forest respectively. These results mirror the observation made by Ricciato, Coluccia (2023), underscoring the critical importance of the initial data's distribution.

An alternative research direction to the more classical estimation methods in the previous papers is the usage of deep learning to facilitate estimating the population. Zong et al. (2019) present a deep neural network called DeepDPM, which generates higher-resolution population mappings from coarse data using Point of Interest (PoI) data divided into four distinct classes. The collected mobile data is assigned to a regular grid using Voronoi polygons, the PoI data is mapped to the same grid. Their model generates a full time-series of the dynamic population at the higher resolution. They conclude that their model works well for static and dynamic population estimation (Zong et al., 2019). This study shows that it is possible to enhance the spatial resolution of the initial training data using ancillary information, thereby lessening the dependence on high-resolution census data.

Another deep learning method is presented by Di Zhu et al. (2020). They describe a method of using conditional generative adversarial neural networks for the process of spatial interpolation and compared it with inverse distance weighted and ordinary kriging methods. Their model outperforms the other two methods and they conclude that deep learning can be used to spatially locate data. This paper showcases the potential of deep learning to spatially distribute the mobile data, and further underlining the importance of the pre-processing steps.

A new approach is also introduced by Cecaj et al. in 2020. They employ deep learning techniques on mobile phone data, albeit with the focus not on spatial data localization but rather on forecasting cell-based time series. Their data is active mobile phones in a 250m by 250m raster in a four month time span. The initial spatial location of their data was calculated by the network provider supplying the data. They compared the time-forecasting methods autoregression, seasonal average, ARIMA, Prophet and Holt-Winters Exponential smoothing with multilayer perceptron, convolutional neural networks, long short-term memory networks and a Hybrid Cnn-Lstm Model. They conclude that deep-learning methods are easier to apply on time-forecasting and offer similar results to statistical methods. They also note that their model does not necessitate large amounts of feature-engineering or even any expertise in the domain of mobile networks (Cecaj et al., 2020). This paper contradicts previous insights that ancillary data should be used to improve model results, but it needs to be restated that they did not conduct the initial data distribution, which still may benefit from supplementary data.

The key takeaway of this chapter is that currently there is no standardized method-

ology, and there is no consensus on the exact kind of mobile network data needed to estimate the population with the highest accuracy. Also, studies usually only pertain to dense urban areas, and do not apply their methods to more general, country-wide or global use cases.

4 Case Study - Predicting Munich's Dynamic Population

In this section, the paper presents a case study that explores the potential of mobile network data for predicting nighttime and dynamic populations, building upon the current state of research discussed in the previous section. The case study examines the city of Munich and the surrounding counties and is based on the method tried by Khodabandelou et al. in 2019b, but it differs in several key areas. First, it uses a different data basis than Khodabandelou et al. and most other studies, because it uses the raw performance data on network cell level without any method of linking different network events like calls to an identifying ID. The data is thus completely non-identifiable. This hugely simplifies the data acquisition process and protects the user's privacy. This paper also uses best server raster data for each frequency band to accurately determine network cell coverage instead of rougher approximations like Voronoi or Thiessen polygons. There is also no preceding determination of land use using spatial clustering with the Median Week Signature (MWS) method as described in Furno et al. (2017). To replace this, a modified CORINE (Coordination of Information on the Environment) landcover data-set is used to spatially divide the study area. Another important difference is that the data is not spatially interpolated onto administrative units, but on a 100m by 100m INSPIRE-conform raster, which allows for a higher resolution and an easier comparison with the ground truth census data in the same format. At last, the study area does not only include urban city areas but a total area of $4696km^2$, which includes suburban and rural areas as well as large agricultural and forested areas.

The following subsection explains the used data set in detail and describes the uncertainties. The subsequent part describes the used methodology for the case study.

4.1 Data

In order to deliver an accurate model, this paper uses several data sets from different sources and years. Some of the data, such as the building and best server data is used in the pre-processing step. This data, together with the mobile phone metrics was provided by the Telefónica Germany GmbH & Co. OHG and is not publicly available. The 2011 census data is used to train the population estimation model and the remaining census data is used as validation for the model results. These data sets are:

- Four different mobile phone data metrics
- Best-server data for each frequency band
- Building polygon data
- German census data from 2011
- Bavarian county-level census data from 2021
- Munich census data from 2022
- CORINE landcover data from 2018

Additionally, administrative boundaries were sourced from the German Federal Agency for Kartography and Geodesy¹⁰. The underlying map used in some figures is OpenStreetMap by the OpenStreetMap Foundation¹¹.

4.1.1 Mobile Phone Data

The employed mobile phone data comprises performance metrics obtained from all operational 4G network cells within the study area. 2G data was not used because of its analog nature, causing distinct differences in defining metrics such as active users and voice traffic compared to the more contemporary digital technologies. Moreover, conversion of the data is not a straightforward process due to these disparities. In addition, 2G also has the lowest amount of users and traffic in Germany. A high amount of devices only using 2G are not mobile phones, but industry machines using Machine to Machine (M2M) communication and are thus not of interest for our analysis. Furthermore, 3G data is not included because it was deactivated in Germany in 2021, thus no data is available. 5G data is already included in the 4G data set, because most modern applications of 5G use the Non-Standalone Architecture (NSA). Here, the 4G network is used for the initial connection between mobile device and cell, and all metrics are also counted for the corresponding 4G antenna (Sauter, 2022:p. 119 f.).

¹⁰<https://gdz.bkg.bund.de/index.php/default/digitale-geodaten/verwaltungsgebiete/verwaltungsgebiete-1-5-000-000-stand-31-12-vg5000-12-31.html>, Accessed 20.03.2023

¹¹<https://www.openstreetmap.org/>, Accessed 21.04.2023

On top of that, there is a small amount of indoor network cells in the study area. These cells are usually found on the inside of big buildings like convention centers, arenas, airports or main train stations and other infrastructure hubs or inside of street tunnels. However, these cells do not have a corresponding best-server coverage area. Because of this, the values of each small cell are added to the network cell with the smallest covered area overlaying the position of the small cell, as described in Ricciato et al. (2020). While the number of indoor network cells is small, they often bear the brunt of data traffic in areas with exceptionally high population densities. Therefore, their inclusion is important for achieving an accurate population estimation.

Additionally, the data set contains no personally identifiable information and is unobjectionable from a data protection perspective. The raw data is divided into the six different frequency bands used by the Telefónica network: 700 MHz, 800 MHz, 900 MHz, 1800 MHz, 2100 MHz and 2600 MHz. Further, the polling rate is one hour and during the study period new cells were added and others removed, which necessitated using different best server raster for different months. Another reason for missing values are technical problems,

The included metrics are active sim-cards, voice traffic, voice calls and data traffic. These metrics are defined in further detail in the following sections.

Active Users

This metric measures the number of SIM-Card activations in one hour. An activation occurs whenever a device with an active Broadband receiver and SIM-card enters the cell-area. It consists of a short relay of information between device and cell. No identifying information of these SIM-cards is used, thus one SIM-card could be activated multiple times by, for example, exiting and re-entering into the same cell multiple times. But simply staying in one cell only counts as one activation per polling period. Nevertheless, even if not moving a device can be counted in multiple overlapping cells if it switches between them.

Voice Traffic in Erlang

This metric measures the aggregated call duration in each cell. This includes outgoing and incoming calls. One unit of Erlang is one hour of incoming or outgoing voice traffic, divided into any amount of different calls. It is important to note that voice calls made with Voice over Internet (VoIP) protocols are not included, as they can't be measured by the network provider.

Voice Calls

This metric measured the aggregated number of successful outgoing calls. It should be noted that outgoing calls that were interrupted before establishing a connection because the caller hung up or because of technical problems are not counted. Additionally, if a person moves from one cell to another while on a call, the call is counted for both cells.

Data Traffic in GB

This metric measures the aggregated data volume transferred in this cell in one hour. Here, upload and download are both counted. Also, it is measured in GB with two decimal places, so down to 10 MB.

4.1.2 Best-Server Data

There is a Best-server raster for each of the six frequencies 700 MHz, 800 MHz, 900 MHz, 1800 MHz, 2100 MHz and 2600 MHz. The rasters are in an 100m by 100m INSPIRE conform grid. INSPIRE, or infrastructure for spatial information in Europe, is a directive of the European Union to create a unified spatial data infrastructure, facilitating easier data sharing. One aspect of this directive are guidelines on the structure of raster data¹². Therefore, all INSPIRE-conform raster grids are compatible with each other. Here, each grid cell is assigned to the network cell with the strongest signal at its position for each frequency. Also, there are no overlaps in each raster, but the best server areas are often non-contiguous. Because new network cells were added or removed, different best server raster were used for data from different months.

4.1.3 Census Data

This paper uses three different census data sets. The data used to train the model and to predict dynamic and night population is from the 2011 register-based census in Germany, because it has the highest spatial resolution (100m by 100m). However, a register based census surveys only a part of the total population. The census data additionally relies heavily on population registers¹³. It is known that the data about address and general location of any individual can be taken from the registry, whereas additional information like educational background, profession and housing situation can't be easily inferred. For this, supplementary surveys were conducted.¹⁴ The used census data is in the same INSPIRE conform grid as the

¹²<https://inspire.ec.europa.eu/about-inspire/563>, Accessed: 04.04.2023

¹³https://cros-legacy.ec.europa.eu/content/register-based-census_en, Accessed 22.04.2023

¹⁴https://www.zensus2011.de/DE/Zensus2011/Methode/Methode_node.html, Accessed 23.04.2023

Bestserver and CORINE land cover data. To be specific, it consists of one raster, where the value of each raster cell represents the permanent residents of this specific cell. In addition, to ensure privacy, cells containing only one inhabitant are assigned a value of zero, while cells with two inhabitants are assigned a value of three¹⁵.

A newer census data set is the population per county in Bavaria, collected in 2021¹⁶. For the city of Munich, a 2022 data set with the population per city district was used¹⁷. These two data sets are on a lower spatial resolution than the 2011 census, and thus can not be used to train the model, but they can be used to validate the model results on an aggregated county and city district level and to compare the population of the districts and counties.

4.1.4 Building Footprint Data

The used building layer is a vector layer with one polygon for each building in the study area from 2022. There is no additional metadata about building type or similar attributes. The data was bought from the Nexiga GmbH and is not publicly available

4.1.5 CORINE Land Cover Data

The CORINE land cover data provides detailed land cover information for the European continent. It is maintained by the European Environment Agency of the European Union. To be precise, it uses a standardized classification system, containing 43 distinct classes. Also, the data can be freely accessed and is available as a INSPIRE conform 100m by 100m grid raster or as a vector layer. This paper uses the latest raster data from 2018.¹⁸

4.1.6 Uncertainties in the Data

It is known that there are always uncertainties in using real-world data. At first there is the problem of dis-aggregating the data from a network cell to all raster cells in its area. There is a need to use additional data at this step to better distribute the data. However, this can introduce new biases. For example, using a building

¹⁵<https://www.zensus2011.de/DE/Home/Aktuelles/DemografischeGrunddaten.html>, Accessed 23.04.2023

¹⁶https://www.statistik.bayern.de/produkte/statistik_kommunal/index.html, Accessed: 23.07.2023

¹⁷<https://stadt.muenchen.de/infos/statistik-bevoelkerung.html>, Accessed: 07.05.2023

¹⁸<https://www.eea.europa.eu/data-and-maps/data/copernicus-land-monitoring-service-corine>, Accessed 22.04.2023

weighting layer leads to an under-reporting of activity at temporary event spaces, on streets and not built-up areas. That's why without any additional information the data could simply be divided into equal parts depending on the amount of cells. But with this approach densely populated areas will be under-reported, sparsely populated ones over-reported.

A second uncertainty is the best server layer. Mobile network providers usually calculate these layers for their internal network coverage and optimisation planning. Thus, the exact used model is usually not open for research and the used parameters and methods can't be scientifically verified. Two studies using best server data from different providers thus can not be compared easily.

A further uncertainty is that the four different collected metrics can not be directly related to the actual number of persons at a given location. As not every individual has a mobile phone, and not every mobile phone is used by only one person.

Additionally, it is important to acknowledge that the collected active user metric faces the challenge that a single mobile device can be simultaneously counted as an active user in multiple network cells, potentially leading to an overestimation of this metric in the data set. Conversely, the data usage and Erlang metrics do not have this issue.

Another uncertainty is that the data does not encompass the whole German mobile network market. Besides Telefonica, there are two other big network providers with around a third of the mobile customers each. Furthermore, customers of different providers can also have a different social-demographic markup, further complicating the matter.

An additional constraint is the considerable variability in spatial accuracy for the collected network metric. Network cells can vary in size, which means the accuracy in urban areas, where more smaller cells are located, is higher than the accuracy in non-urban areas (Spinney, 2003).

The indoor cells represent an additional factor of uncertainty. These are network cells only found inside of buildings. Their actual network coverage is difficult to calculate. But usually they do not only extend inside of their building. Therefore, spatially assigning the values measured by these cells is still an open question. Besides, any spatial distribution of the measurements from these cells would need to differentiate between indoor cells in large buildings and in underground structures such as tunnels or train stations.

Furthermore, the omission of 5G and 2G data introduces an additional layer of uncertainty. The 2G data is not included because it is not comparable due to its

analog nature. In spite of that, the exclusion of this data can introduce biases, for example that different demographics are more likely to have older phones without 4G/5G access and would not be included in the data. At the same time, the 5G data is indirectly included by adding their metrics to the 4G Antenna that establishes the initial connection. But this does not consider the different coverage area of the 5G Antenna. It has to be considered that the initial connection of a short-range 3600 MHz 5G Antenna could be established by a long-range 700 MHz 4G Antenna. The measured data would then be added to the 4G Antenna that has a different coverage area, making a correct spatial interpolation of the collected data impossible.

4.1.7 Study Area - Munich

The study area encompasses the Bavarian counties of Munich city, county of Munich, Fürstenfeldbruck, Dachau, Freising, Erding, Ebersberg and Starnberg. Figure 6 shows the area marked in green. The black rectangle indicates the extent of the raster that was used for most calculations. This raster is big enough to contain all network cells that overlap the study area. The different counties are marked in black. All network cells with a best-server area intersecting the raster extent were taken into account. This means that some network sites with locations outside of the extent where also taken into account.

County	Census 2021	Area in km^2	Population per km^2	Population per 100 m cell
City of Munich:	1,487,708	310.7	4,788	48
County of Munich:	349,837	664.25	527	5.3
Fürstenfeldbruck:	218,579	434.8	503	5
Freising:	181,144	799.85	226	2.3
Dachau:	155,449	579.16	268	2.7
Ebersberg:	144,562	549.39	263	2.6
Erding:	139,622	870.74	160	1.6
Starnberg:	136,747	487.71	280	2.8
In total:	2,813,648	4696.6	599	6

Table 1: Population values, area, population density, and density for the 100m by 100m cells in the different counties of the study area in the year 2021.

To illustrate variations in population, Table 1 presents an ordered listing of total population, area, population density per km^2 , and population density per raster cell, sorted in descending order by total population. The study area had a total population of around three million permanent residents in 2021. With its area of $4,696.6 km^2$ it the population density is 599 inhabitants per km^2 . The plurality of the population, around 1.5 million is located in the city of Munich.

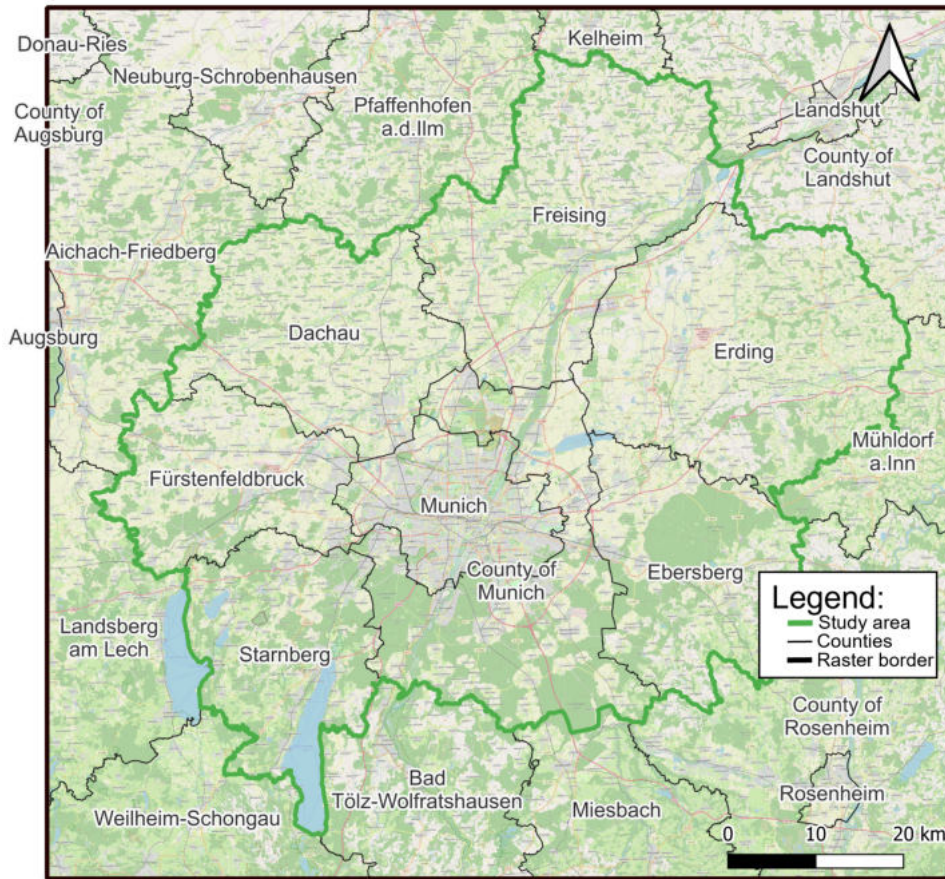


Figure 6: Overview of the study area

4.2 Methodology

The methodology used in this thesis can be divided into three main parts. The second and third part are taken from Khodabandelou et al. (2019b). The first step, pre-processing using building weighting is taken from Aasa et al. (2021). The original first step of dynamically classifying the data into clusters with different mobile phone usage from Khodabandelou et al. (2019b) was not used because it would go beyond the scope of this paper. It is replaced by using the CORINE land cover data. The three parts, as also shown in more detail in the flow chart in figure 7, are:

1. Pre-processing the raw network cell performance data using ancillary data
2. Building a linear regression model to predict night population
3. Building a multivariate model to predict the dynamic populations

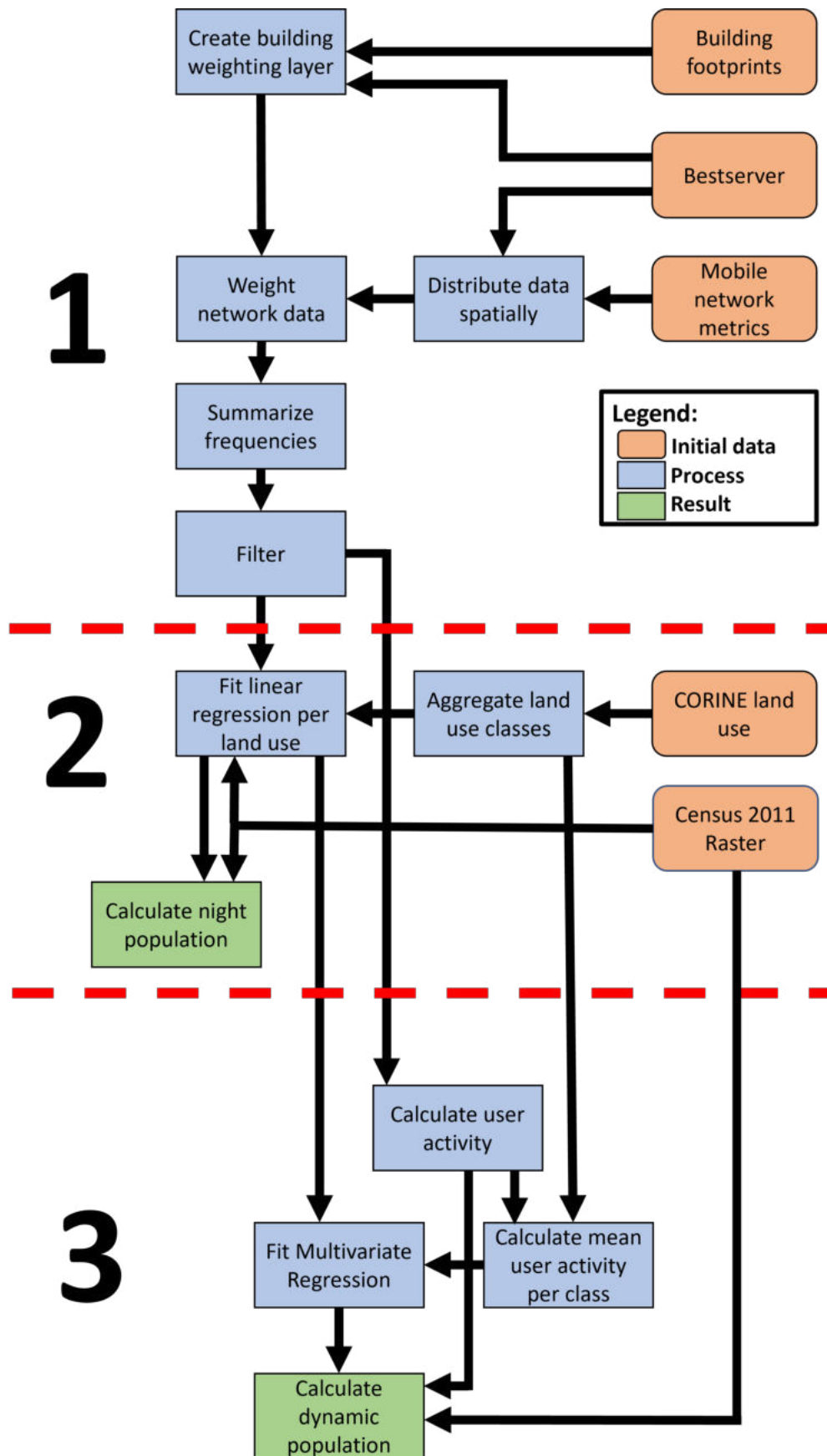


Figure 7: Schematic of the three step process: 1. Data preparation, 2. Night population estimation, 3. Dynamic population estimation

At first, the building weighting layer is created, which is used to better distribute the data over the study area. Then this data is used together with the raw performance data and the best server raster of each frequency to create one raster for each hour and each of the examined metrics. This data is then filtered, selecting only the metric and weeks with the highest correlation. After that, the data is used to refine the used land use classes in the CORINE land use raster. Subsequently, for each of the different land use classes, a separate linear regression model is fitted. The RANSAC regressor used for this purpose detects outliers and discards them before performing the fitting. The resulting intercept and coefficient variables α and β from formula 1 in chapter 2.6 can then be used to predict the night-population for each raster cell.

For the third part, a measure of the proportion between user activity and user count is calculated. From this, a average activity value is calculated per land use class. This average is used with the variables α and β from the night population estimation to fit a second linear model, creating four new parameters that are then used in the multivariate model for predicting the dynamic population. As a result, this model allows to calculate the predicted population for each hour on each weekday for each cell depending on the activity measure.

From a technical perspective, python 3.11 was used together with numpy 1.25.0 and pandas 2.0.3 for general data reading and transformation tasks. For vector data sets, geopandas 0.13 was used and for raster data rasterio 1.3. The regressor models from the sklearn package were used to fit the models. The creation of the building weighting layer was accomplished using ArcGis Pro 3.1.1 from ESRI. The complete python code for the initial data pre-processing, model training and validation is accessible under <https://git.rz.uni-augsburg.de/vesterda/master-thesis-daniel-vester>. For data-protection and licence concerns, none of the used network performance, coverage and building data is included.

4.2.1 Preparing the Building Weighting Layer

The first step in preparing the data for the model is creating a weighting layer. This layer will be used to improve the spatial interpolation of the measured values. The method described in this section follows the approach of the housing weighted interpolation (HWI) described by Aasa et al. (2021). But unlike their study, actual best server data is used to determine cell coverage instead of Thiessen or Voronoi polygons, which is congruent with Ricciato et al., who assess that using real-world actual coverage data can be superior to simple Voronoi approaches (Ricciato et al., 2020).

The creation of the weighting layer starts with a polygon layer for the whole study area, with one polygon for each building. There are no overlapping polygons. In the first step, this vector layer is rasterized to raster F with $10\text{m} \times 10\text{m}$ cells. For this the rasterize function from the gdal translator library was used ¹⁹. This function assigns each cell the value 100 if it's center point is inside of a building polygon, if not it is assigned the value 0. The resolution of 100m^2 is chosen to maintain a balance between precision and computational cost. An example image section from downtown Munich for this raster can be seen in figure 8. It shows that in dense urban areas streets, open spaces and built-up areas are clearly discernable at this resolution.



Figure 8: An example of the 10m by 10m building raster displaying downtown Munich

The next step is aggregating this higher resolution raster to a coarser 100m by 100m INSPIRE-conform raster B. For this, each cell of raster B is assigned the sum of all cells in raster F that are encompassed by the B cell. The aggregate tool from ArcGis Pro was used for this purpose ²⁰. This raster B now shows the approximated total building footprint in square meters in each cell.

The next step is calculating the total building footprint that each network cell covers. For this, the best server data T^{freq} for each frequency band is used. This data includes all network cells k and their corresponding best-server coverage. Six new

¹⁹https://gdal.org/programs/gdal_rasterize.html, Accessed 14.06.2023

²⁰<https://pro.arcgis.com/en/pro-app/latest/tool-reference/spatial-analyst/aggregate.html>, Accessed: 14.06.2023

raster C^{freq} are created by adding the square meter values B_i of all raster cells belonging to the best coverage of a network cell.

The last step is creating the weighting layer. This is accomplished by dividing each cell i of the building layer B through each of the cell-wise footprint rasters C^{freq} , as shown in equation 3. The cell B_i is divided through the value of the network cell C_k^{freq} that it is contained in. This equates in raster cell W_i^{freq} . There is no network cell with a building footprint of zero. This process allocates the share of each raster cell's building area to the total building footprint in the corresponding network cell. This results in one weighting raster W^{freq} for each of the six used frequencies.

$$W_i^{freq} = \frac{B_i}{C_k^{freq}} \quad (3)$$

The core notion driving the implementation of this weighting layer revolves around the fact that the majority of individuals reside in buildings, and these buildings tend to be concentrated in regions with denser populations. Consequently, it is plausible to view buildings as a reliable indicator of how the population is distributed across an area (Aasa et al., 2021).

4.2.2 Spatial Interpolation of the Raw Data

To distribute the data from the coarse network-cell coverage level to the 100m by 100m grid cell level, a dasymetric approach is chosen. This approach distributes the data based on the prepared ancillary building weighting layer from the previous step. The used raster format has the advantage of fast calculations of stacks of multiple raster and easier aggregation of data at the same spatial position. It is also more robust against changes in the best server areas and has a higher stability over time. The raster approach allows automatic inclusion of changes without recalculating the entire layer, making it suitable for dynamic data updates. However, it can lead to increased computational complexity due to a higher number of features compared to Voronoi/Thiessen polygons. Another advantage of rasters is that there are no public administrative subdivisions in Germany with a high spatial resolution. Administrative subdivisions also vary between the different German states. Therefore, by using rasters, the desired spatial resolution can be freely used.

The raw data is provided in tabular form, with one table for each week of data for each metric. In a first step, the tabular data for each hour is joined with the six best server raster for each of the frequencies. Each raster cell is assigned the value the corresponding network cell had at his hour. This creates a total of 4032 $V_{metric,day,hour}^{freq}$ rasters per week. To make the selection of specific weekdays or hours

of the day easier, the time of the raster is identified by a combination of day-hour. Then these rasters $V_{metric,day,hour}^{freq}$ are weighted by multiplying them with the corresponding building weights layers W^{freq} . With this method, all cells receive their measured value times their proportion of the building footprint in their network cell. A raster cell containing 50% of the building area would thus also receive 50% of the measured values. It follows that raster cells without any building area are assigned a value of 0. Because of this, populations far away from any buildings or in temporary buildings can not be estimated with the methodology in this paper.

In the next step, all six different frequency rasters for each *metric, day, hour* combination are summed up to create the raster $S_{metric,day,hour}$. Each individual raster cell $V_{metric,day,hour,i}^{freq}$ is multiplied by the corresponding weighting value from W_i^{freq} . Then the values of all raster cells at the same position *i* are summed over all frequencies. Equation 4 shows this procedure together with the weighting. This methodology creates 672 rasters for each week of the study period (168 hours times 4 metrics).

$$S_{metric,day,hour,i} = \sum_{freq} V_{metric,day,hour,i}^{freq} * W_i^{freq} \quad (4)$$

The spatial interpolation step is the most computationally expensive process in this methodology, which necessitates saving the pre-processing results in a permanent format. The following steps can then be performed with a large variety of different inputs, allowing for easy comparison and evaluation of the best performing input data.

4.2.3 Metadata and Time Filtering

In this chapter the process on deciding on which metric is the most suitable to predict the night and dynamic population is explained. A first step is examining the connection between the different metrics. Figure 9 illustrates the relationship between user amount and the metrics of data usage, call volume and Erlang. All values are in logarithmic scale and are exemplary median values from Wednesday 12 am. It also shows the corresponding pearson correlation coefficients.

As can be seen, these metrics are highly correlated, with only a low amount of outliers. There is one main group of outliers at the bottom left, with lower values for data usage, Erlang or calls.

The pearson correlation is also high, with values between 0.94 for the Erlang-User correlation and 0.97 for Calls and Users.

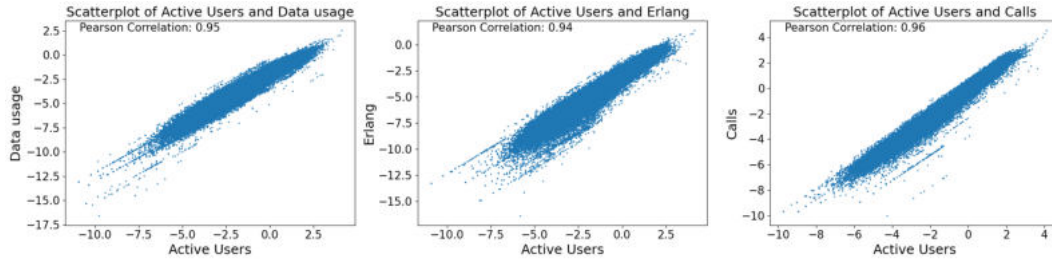


Figure 9: Relationship between active users and data traffic, call volume and Erlang values and the corresponding pearson correlation coefficient

Intuition dictates that there is a close relationship between the number of calls and the total call duration. This relationship is shown in figure 10. The pearson correlation coefficient is included, and all data is in logarithmic scale. As can be seen, there is a clear linear trend, with a few small clusters of outliers.

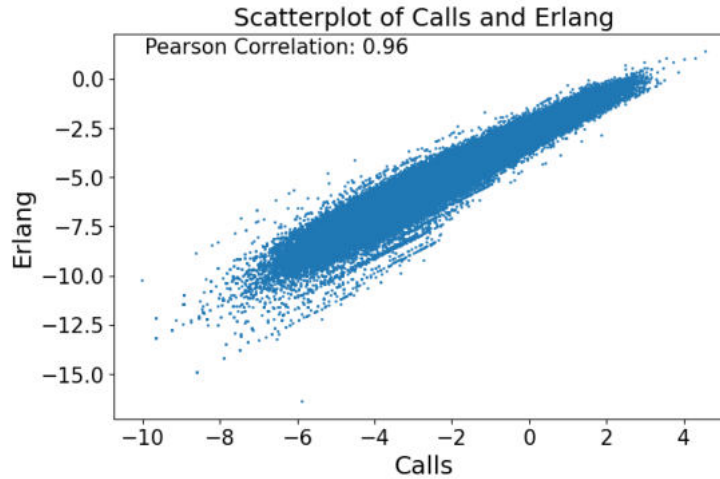


Figure 10: Scatterplot of number of calls and Erlang and the pearson correlation coefficient

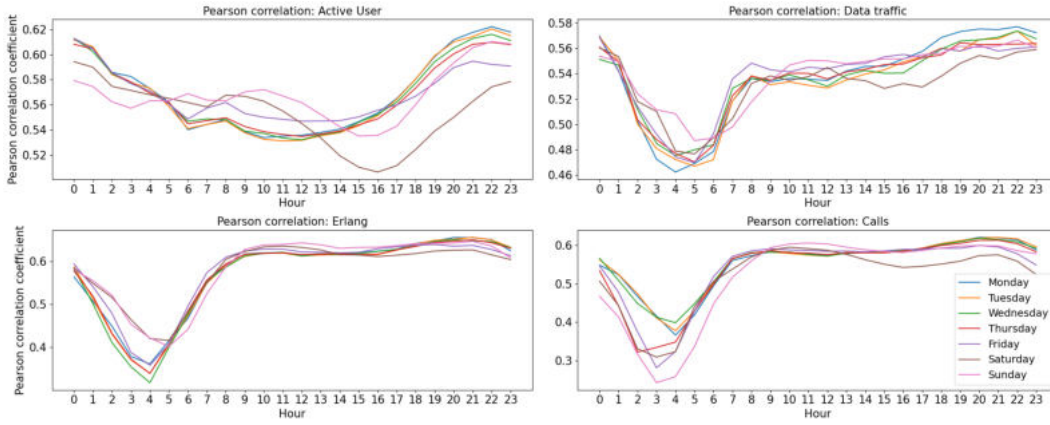
The pearson test also shows a high correlation between these values, with a pearson correlation of about 0.97. This finding aligns with Kang et al. (2012), where they also find high correlations between these metrics. In their study they conclude that call volume is better suited for population estimation than Erlang values which does match the results shown in figure 9.

It can be said that all metrics should produce similar results because of their high correlation between each other, but to find the most suitable metric, a closer look at their correlation with the underlying census data is needed. This decision is also dependent on the used time of day, as different metrics have a different amount of correlation at different hours.

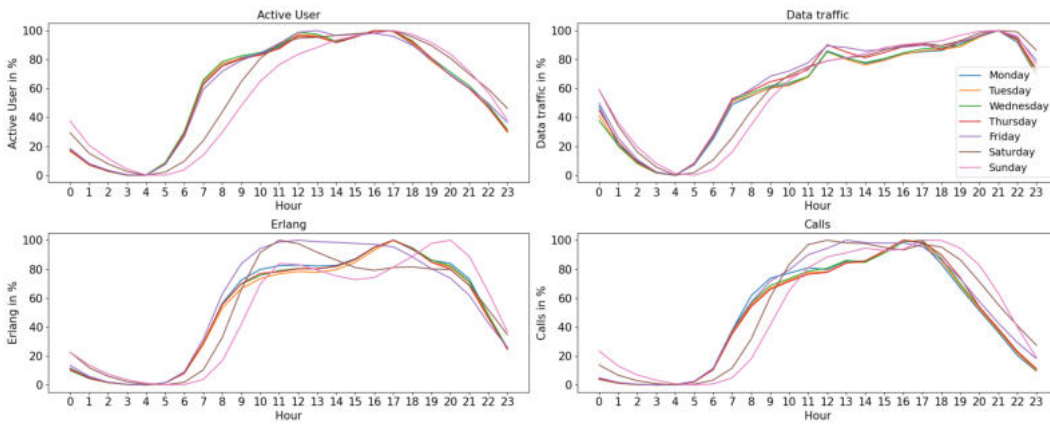
One dimension which can significantly change the correlation between ground truth and metric is time. The time of day and the weekday have a big impact on correlation as can be seen in figure 11a. The plots in figure 11a show the pearson correlation coefficient between the census data and each of the four metrics for each hour of each day of a median week. The first metric, ranging from a pearson correlation coefficient of about 0.48 to 0.62 is the amount of active users. There are stark differences between the days, with Saturday having the lowest correlation after 12 am, and Sunday having a completely different course with two small peaks before 4 pm as opposed to the other days. The five working days have similar courses, with the highest correlation at around 10 pm, and the lowest at 12 am. Monday at 10 pm has the highest correlation at around 0,62.

The three other metrics, data traffic, amount of voice calls and duration of voice calls all have their lowest correlation at around 4 am in the night. Data traffic peaks at around 10 pm in the evening at 0.58. The lowest point is at 0.49. The different days have a similar course up until 10 am in the morning, where the Sunday data has the highest correlation up to 6 pm in the evening. The working days have a local minimum at 12 am. The voice call duration has two small peaks at 11 am and 8 pm, where the 8 pm peak is higher at around 0.65. The lowest point, at around 4 am, is at 0.33 on Wednesday. All days have a similar course, with Sundays and Saturdays having a higher lowest point at 4 am at around 0.4. The voice call amount ranges from 0.24 for Sundays at 4 am in the night to 0.62 on Tuesday at 7 pm in the evening. The different days differ mostly at night, between 2 am and 6 am. Monday, Tuesday and Wednesday show the lowest correlation of about 0.42, whereas the rest of the days are lower.

The plots in figure 11b show the summed absolute values of each metric over the course of each day. For data protection reasons, the actual numbers can not be shown, thus the y-Axis is normalized to a scale between 0 and 100%. Comparing the pearson coefficient to the absolute numbers shows that a higher usage does not necessarily lead to a higher correlation. One exception is 4 am for the data traffic, Erlang and calls metrics. Here the pearson correlation and the usage is at it's lowest point. Interestingly, the active users metrics has a course more similar to the Erlang and calls metrics, and is not similar to its pearson correlation graph. To sum it up, these four graphs show that most users are active at 11 am and 16 pm on all workdays, and the least are online at around 3 to 5 am. In addition, for data usage, the lowest point is at 4 am, a small peak at 12 am and its highest point is at 10 pm. Erlang and calls show a similar pattern for Monday trough Thursday, with their lowest point at 4 am, a plateau between 9 am and 1 pm, and a peak around 5 pm. On Friday to Sunday, the usage stays on one level from 9 am to 4 pm.



(a) Pearson correlation for all metrics and each day of a median week



(b) Relative activity values for the whole study area for all metrics of a median week. Percentages are relative to maximum of each metric

Figure 11: Pearson correlation (top) and relative activity values (bottom) for each metric during a median week

Table 2 shows additionally the highest pearson correlation for each of the four metrics, together with corresponding weekday and hour. The highest correlation is marked in red.

Metric	Weekday	Hour	Pearson correlation coefficient
Active Users	Monday	11 pm	0.622
Data traffic	Monday	11 pm	0.577
Erlang	Tuesday	10 pm	0.655
Calls	Monday	9 pm	0.621

Table 2: Maximum pearson correlation, rounded to three digits after the decimal point, for each of the four metrics

To summarize, the highest pearson correlation between the census data and the collected data is for the Erlang metric on a Tuesday at 10 pm. This correlation does not corroborate with the relevant literature, where the time between 3 am and 5 am in the morning for the voice call amount is recommended, as in Khodabandelou, Gauthier, Fiore, El-Yacoubi (2019b). Because of this contrast, on one hand a different night population model will be calculated with data from each hour of each day for each metric, and then a comparison of these models will evaluate which one has the best results. On the other hand, a model fitted with the Erlang measurements from Tuesday at 10 pm will also be used for further examinations and comparisons between the different land use classes.

4.2.4 Week Filtering

One open question is which weekly data should be considered for the static analysis as dynamic populations can vary substantially, with different daily, weekly and yearly patterns. As the data is collected in weekly intervals, and the dynamic population models needs to be able to predict the population on all days in a week, it is prudent to select whole weeks of the data and not smaller increments. The different days of a week can also vary depending on workdays, weekends, public and school holidays and similar events. The calendar in figure 12 shows public holidays, that means days where most shops, educational institutions, and businesses are closed, school holidays and weekends. The spatial patterns of school children should be significantly different on these days, because they do not congregate at school. As can be seen, it is hard to find time spans longer than 42 days without any holidays. But it is not required to choose consecutive weeks.

For this paper, four weeks from the whole data set were chosen. These weeks were chosen because they do not include any school holidays or holidays.

- **Week 16:** 17.04. - 23.04.
- **Week 21:** 22.05. - 28.05.
- **Week 25:** 19.06. - 25.06.
- **Week 26:** 26.06. - 02.07.

This selection over the span of four months attempts to reduce the effects of seasonality. It also includes the weeks with the lowest amount of obvious measurement errors, network outages and missing data.

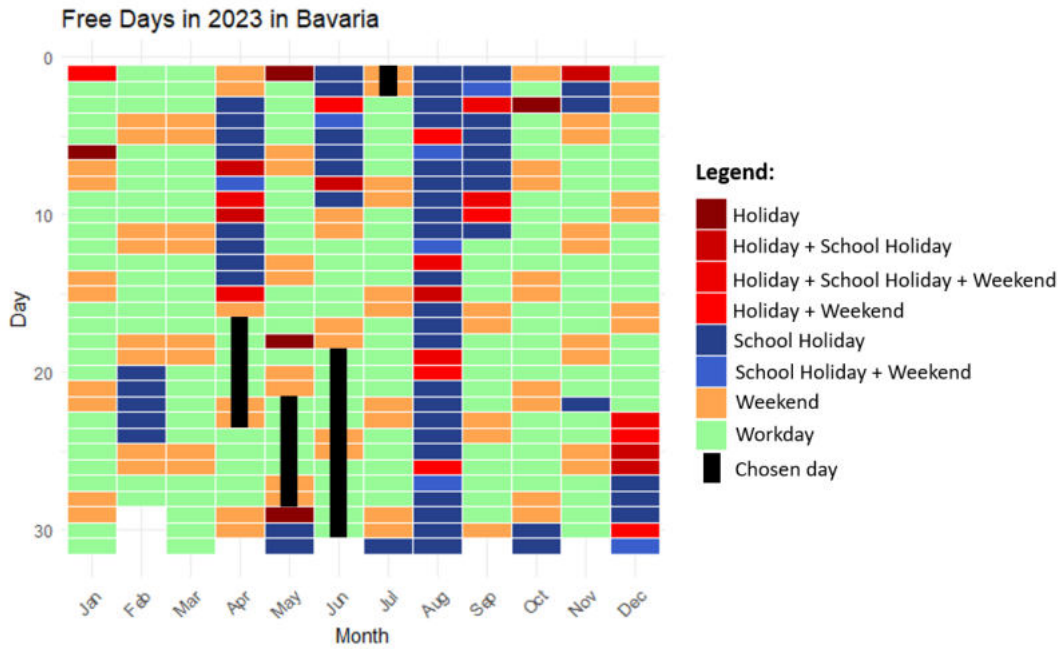


Figure 12: Holidays and school holidays in Bavaria in 2023

4.2.5 Deciding on Selected Land Use Classes

The following night population regression model is run separately for different land use classes. In this paper, CORINE land cover is used for this purpose. CORINE is comprised of 44 distinct land use classes. Because this thesis is focused on areas with human population, CORINE classes for non urban land uses were summarized to more general classes. The Agriculture class is comprised of CORINE classes 211-244, Forest consists of the classes 311-324, Barren Land of 325-335, Water of 411-523 and the new Open pit class consists of 131 - Mineral extraction sites and 132 - Dump sites. Also the classes 122, 123 and 124 were combined to a transportation class containing roads, rails, ports and airports.

This results in 12 classes, which are listed in table 3 with their corresponding count of raster cells and the median amount of active users in each class over the span of a week. The table also includes the median active users per raster cell in each class, and is ordered by this value. The highest value in the active user and cell count columns is marked in red. As can be seen, the biggest class is agriculture, followed by forests and then discontinuous urban. The smallest class is construction sites, then open pits and green urban. For the user amount, discontinuous urban has the highest value with 66778 active users, followed by agriculture and then industrial/commercial and continuous urban. The lowest user amount can be found for barren land, and then construction sites and open pits. The continuous urban

class is the only class with higher user counts than cell counts, because it is the most densely packed and populated class. These 12 classes vary considerably in size and user amount, which could mean that some similar and smaller classes could be combined.

Class	Active users	Cell count	User per cell
Continuous urban	9175	2249	4.08
Industrial/commercial	13473	14269	0.944
Discontinuous urban	66778	74517	0.896
Transportation	1559	3974	0.392
Green urban	901	2578	0.349
Sport/Leisure	966	5472	0.177
Construction sites	47	378	0.124
Open pit	58	1296	0.045
Agriculture	18438	636537	0.029
Forests	2162	264929	0.008
Water	100	16211	0.006
Barren Land	7	4392	0.002

Table 3: The 12 classes, and their median active users over a week and count of raster cells per class as well as the median active user count per cell, sorted by user count per cell

The exact number and aggregation of these remaining classes is another open question. Figures 13 and 14 shows the log of the proportion of the median amount of active users per 100m x 100m cell over the span of a week. Once for 12 classes (13) and once for an eight classes case, which is the case used in this thesis (14). The difference between the 12 class case and the 8 class case is that four classes were combined with other classes.

First, the water and forest class were combined to the new Forest/Water class. As can be seen in figure 13, in green and blue, they both have a similar course over the week, with a similar magnitude. It is thus sensible to combine them. The open pit class was added to the agriculture class. They both have a similar proportion of active users per cell and a similar temporal course. Also the open pit class is the second smallest class, which makes it sensible to add to a bigger class. Sports/Leisure facilities were added to the green urban class. The sports class has a lower users per cell amount, but their temporal course is relatively similar with two small peaks on the workdays, and one smooth peak on weekends. Also both classes are usually either in urban areas or adjacent to them. Sports facilities also include green areas like football fields which can fulfill similar needs as urban green areas. It can also be hard to distinguish these two classes with, for example, satellite images, because they can be so similar.

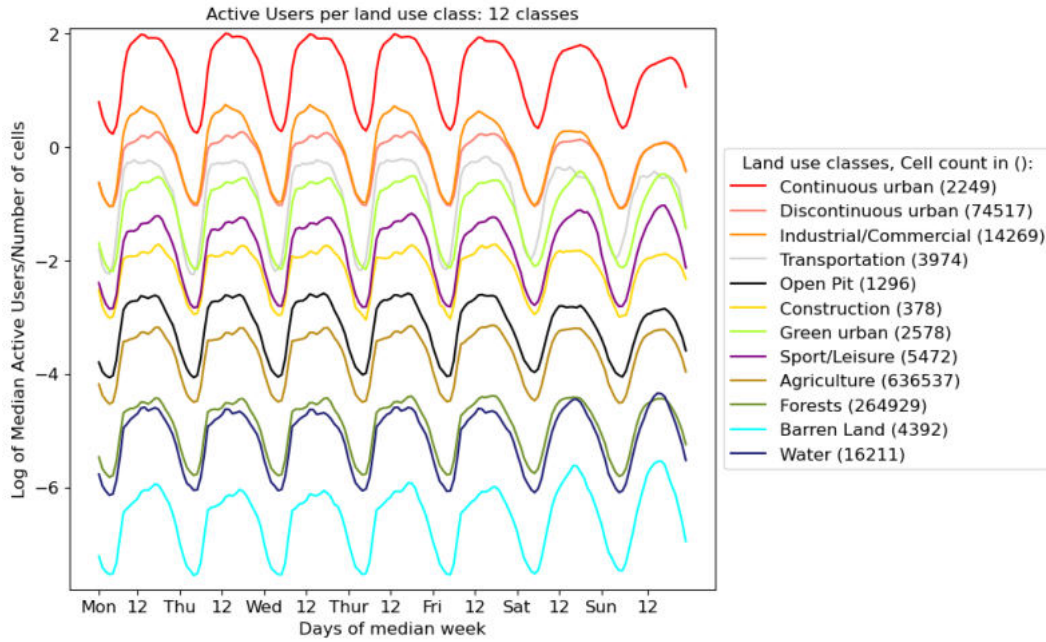


Figure 13: Log of proportion of median active users to cell count with 12 classes

Additionally, construction sites were added to the discontinuous urban class. They should not be their own category for our purposes, because they are not permanent. The CORINE data is from 2018, which means that a high percentage of these construction sites are probably finished, and should belong to one of the other classes. They are added to the discontinuous urban class, because they have a similar course over the span of the week with two small peaks on the workdays, and a flat peak on the weekends as can be seen in figure 13. The transportation class was not added to the green urban class, despite their similar magnitude. This is because this class has higher values on the workdays, and lower ones on the weekends. The green urban class is more homogeneous over the span of a week, with a little more active users on the weekends. Similarly, the industrial/commercial and discontinuous urban classes were not combined, because the industrial/commercial class has much higher peaks on the workdays, as can be seen in figure 14 on the right.

Furthermore, barren land, forests/water, agriculture are non-urban classes with generally a low amount of users. These three classes all have the two-peak pattern on Monday through Thursday, and a more smooth peak on Friday to Sunday. Barren land is the only class of these three, where there are more users on the weekend, than in the workweek. The other have a similar amount through the whole week.

The urban classes are, ordered by user count per cell: continuous urban, industrial/commercial, discontinuous urban fabric, transportation and green urban areas. They all vary more over the course of the week than the lower three classes.

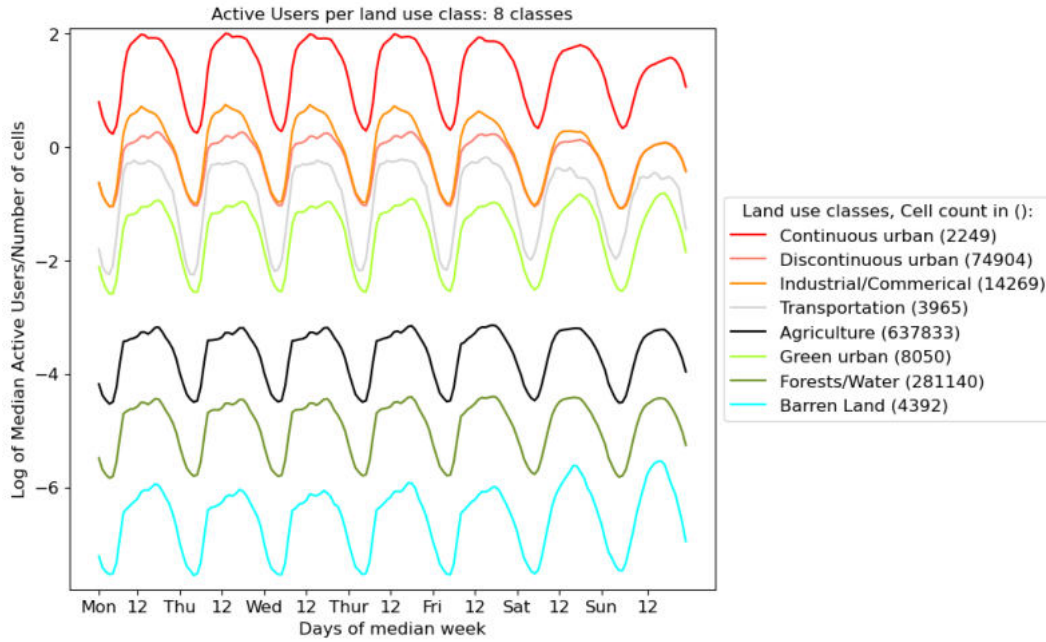


Figure 14: Log of proportion of median active users to cell count with 8 classes

Class	Active users	Cell count	User per cell
Continuous urban	9175	2249	4.08
Industrial/Commercial	13473	14269	0.944
Discontinuous urban	66825	74895	0.892
Transportation	1559	3974	0.392
Green urban	1867	8050	0.231
Agriculture	18496	637833	0.029
Forests/Water	2262	281140	0.008
Barren Land	7	4392	0.002

Table 4: The used 8 classes, and their median active users over a week and count of raster cells per class as well as the median active user count per cell, sorted by user count per cell

Continuous urban has the highest amount of active users, and is unique in the aspect that all peaks are flat. Industrial/Commercial areas have a unique, sharp peak on the workdays, whereas it's peak is more flat and much lower on the weekends. Discontinuous urban, transportation and green urban all have the same two-peak pattern on Monday through Thursday and differ more on the weekends. The first, smaller peak, could indicate the midday break around 11 am to 1 pm for most workers and schoolchildren. The second peak is at the after-work-hours. The weekends only show one peak, at 12 am, the reason for this peak being unclear, as a majority of the population does not work on weekends and thus does not take a lunch break. The Discontinuous urban and transportation classes have less users on the weekend,

whereas urban green spaces have more. This could indicate that green spaces are used for recreational purposes on the weekends.

In the end, each of the 100m by 100m squares is assigned to one land use class, there are no gaps in the data. As discussed earlier, a linear regression model will be calculated for each of these classes separately, thus creating a unique prediction function for each.

4.2.6 Outlier Detection

A last step in the process of data preparation is the removal of outliers. Because of the large amount of real-world raw data, collected from thousands of different network cells over a large time span, the amount of erroneous data can be quite high. The used Random Sample Consensus (RANSAC) regressor algorithm, introduced by Fischler, Bolles in 1981, is used to fit the linear regression model. The regressor is able to detect outliers in the data and automatically excludes these data points from the regression. This is accomplished by starting with the lowest amount of data points possible to compute a model. It then adds all data points that are within the error tolerance of the instantiated model. If the resulting data set is above a specific threshold, the final model can be calculated. If not, a new initial minimal set of data points is selected (Fischler, Bolles, 1981). If no valid model can be found or a predetermined number of tries is reached, the algorithm terminates. The RANSAC regressor implementation from the `sklearn.linear_model` python plugin was used ²¹.

4.2.7 Predicting the Night Population

The first step in predicting the night population is calculating the median for each raster cell-hour-day combination for the four selected weeks. For example, if data from a four week time-span is selected, for each cell at 12 am on a Friday the median of all values of this specific cell on four Fridays at 12 am is taken. This reduces the effect outliers and seasonality have on the data. The input data set of 672 rasters for each of the four examined metrics of active users, calls, Erlang and data usage is thus reduced to 168 rasters each. This step calculates an exemplary week.

The next step is splitting of a training set where only areas with a population value above zero in the census raster are included in the data. This procedure means that a few cells with a census population of one are ignored, because they are set to a value of zero for privacy reasons.

²¹https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.RANSACRegressor.html, Accessed 12.12.2023

After this, the data is again split into the different land use classes. For each land use class, a separate linear regression model is fitted. The intercept and coefficient values of each model can then be used to estimate the night-population over the whole data set.

Fitting Linear Regression After the preparation of the data, the actual model fitting process can now be carried out. Equation 5 shows the connection between the ground truth census population p_i and the value of the chosen metric σ_i in cell i . α and β are the parameters that are fitted. The RANSAC regressor function of the python package `sklearn.linear_model` is used. This regressor uses the LinearRegression function from the same package ²². The used fitting function is the Ordinary Least Squares (OLS) method.

$$p_i = \alpha \sigma_i^\beta \quad (5)$$

Because the trends between census population and metric are on a log-log scale as described in chapter 4.2.3, equation 5 can be transformed to the linear equation seen in equation 6. In this equation $\log(\alpha)$ is the intercept, and β is the sole coefficient.

$$\log(p_i) = \log(\alpha) + \beta \log(\sigma_i) \quad (6)$$

After fitting, the models are not used to directly estimate the population, but the α and β values are extracted. This results in a table with α and β values for each of the eight classes. One example for α and β values, calculated with Erlang data on a Tuesday at 10 pm are shown in table 5. As can be seen, the α values range from 1.391 for Agriculture to 4.719 for the Continuous urban fabric. The β values range from 0.86 for the Industrial/Commercial land use class to 1.43 for discontinuous urban fabric.

Calculating Night Population With the fitted models, the night population can now be estimated. Equation 7 shows how to calculate the estimated population $\hat{p}$ at cell i with the corresponding α and β values and the chosen metric σ_i

$$\hat{p}_i = \hat{\alpha} \sigma_i^{\hat{\beta}} \quad (7)$$

To calculate the night population for the whole study area $\hat{P}$, two new rasters, Alpha and Beta, are generated by joining the modified land use raster with the corresponding α and β values. The end result are two rasters, where each cell has the value

²²https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression, Accessed: 01.02.2023

Class	α	β
Continuous urban	4.719	1.293
Discontinuous urban	3.529	1.43
Industrial/Commercial	1.685	0.86
Transportation	3.175	1.397
Agriculture	1.391	1.005
Green urban	2.792	1.288
Forests/Water	2.351	1.392
Barren land	1.871	0.879

Table 5: Example for fitted α and β values with Erlang data on a Tuesday at 10 pm

of the α and β of its land-use class. Then the predicted night population for the whole study area can be calculated by using the multiply²³ and add²⁴ functions of the python numpy package to calculate the predicted population for the complete raster in one step.

4.2.8 Predicting Dynamic Populations

Estimating the actual, time-sensitive population at a specific location has different challenges compared to the night population model. Ground truth data is not available, and insights gained by the night model do not necessarily hold true for other time frames. Therefore, the approach used in this paper follows the method developed by Khodabandelou et al. (2019b), which postulates a relationship between user count, user activity and dynamic population. The shown approach tries to predict the number of persons in each spatial unit, at each time on any day.

The first step is calculating new α and β values from an adjusted night population estimation model. These values are then used in the fitting of a new multivariate model. Following that, a measure of user count and user activity is calculated. This measure indicates how active each user is. This model produces four new time and land use class invariant parameters, which can then be used to calculate the population at any hour.

Preparing α and β values As said above, the dynamic population estimation uses fitted α and β values from the night population estimation. At first, the fitted values $\hat{\alpha}$ and $\hat{\beta}$ are calculated for each land use class and each hour of the night of each day. These night hours are selected as they are presumed to correspond to periods when ground truth data is most reliably accurate.

²³<https://numpy.org/doc/stable/reference/generated/numpy.multiply.html>, Accessed 2023.02.22

²⁴<https://numpy.org/doc/stable/reference/generated/numpy.add.html>, Accessed 2023.02.22

There is some discussion about which hours constitute nighttime, Khodabandelou et al. (2019b) use a nine hour time-frame between 12 pm in the evening and 8 am in the morning. This paper uses a different time-frame, from 2 am to 7 am in the morning, because the data indicates that these are the hours with the lowest amount of activity. This data is then used to fit one linear regression night-population model for each of these five hours, for each day of the seven days of the four week study period. This produces 140 fitted $\hat{\alpha}$ and $\hat{\beta}$ values per class, so 1120 each.

Subscriber Presence and Activity Levels After preparing the $\hat{\alpha}$ and $\hat{\beta}$ values, a measure of active user count and activity level (number of calls, Erlang, used mobile data per user) for each cell at each hour of each day of a median week is calculated. As with the night-population, the median for each cell-hour-day combination of four weeks is used to minimize the effect of outliers. Khodabandelou, Gauthier, Fiore, El-Yacoubi (2019b) asses a linear, highly correlated, relationship between these activity values and the fitted $\hat{\alpha}$ and $\hat{\beta}$ values from the previous step.

The activity level λ in cell i can be calculated as shown in equation 8. The events metric $\sigma_i(t)$ at time t is divided through the user count $\mu_i(t)$ at time t . σ can be any of the three metrics data, voice calls and erlang. This per-cell activity value is used in the actual estimation of the dynamic population in the **Calculating the dynamic population** section.

$$\lambda_i(t) = \frac{\sigma_i(t)}{\mu_i(t)} \quad (8)$$

As an example, figure 15 shows the activity level λ of one week per land use class. It shows that the eight classes have different patterns of activity on different days and times of the day. The two most similar classes are the continuous and discontinuous urban fabric classes. On Monday through Thursday they almost have the same activity level, with similar peaks at 11 am and 7 pm. On Friday, there are still two peaks, but with the second peak being lower than the first peak at 12 am. On Saturday and Sunday the continuous urban fabric has an higher activity level, but the general pattern is still similar between the two. Interestingly, there are also two peaks on the weekend. On Saturday, the midday is the more active peak, whereas the peak in the evening is lower. On Sunday, the two peaks have a similar magnitude. Industrial/commercial areas also have a similar pattern to the first two classes, but with a much lower activity level. Green urban areas, transportation and the agriculture classes also have a weak daily pattern, with a small peak at midday and slightly higher peak in the early evening on Monday through Thursday. Friday through Sunday are more homogeneous. Forest/Water and Barren land areas only have a daily pattern of almost no usage in the night hours, and a low,

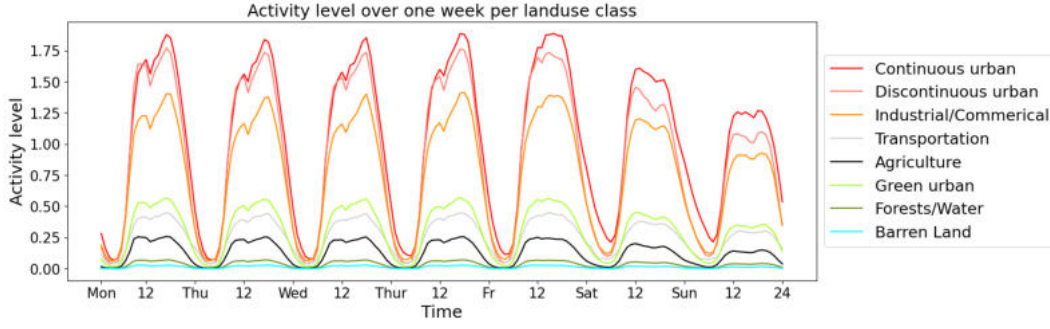


Figure 15: Activity level over the course of one week, per land use class

homogeneous usage in the daytime. Figure 15 shows that there are strong daily and weekly patterns to user activity, which indicates that they could be useful to predict the dynamic population.

To fit the model in the next step, an average activity level per class per time t is needed. This value $\lambda(t)$ for all cells in a class is calculated as shown in equation 9. $\lambda_i(t)$ are all cells in the current class. N is the number of all raster cells in the current class. The average value at time t is calculated by summing the value of all cells in the class at time t , and multiplying it by $\frac{1}{N}$.

$$\lambda^{class}(t) = \frac{1}{N} * \sum_{i=0}^N \lambda_i(t) \quad (9)$$

Whereas Khodabandelou, Gauthier, Fiore, El-Yacoubi (2019b) found a strong linear correlation between these average values and the fitted $\hat{\alpha}$ and $\hat{\beta}$ values, the correlation in this case study is often not strong. Table 6 shows the pearson correlation coefficient between the average activity level per class, the fitted α values on the logarithmic scale and the fitted β values. There are high correlations between the α values of continuous urban fabrics, road, rail, airport and infrastructure and the barren land classes. The α value from green urban areas shows a low to medium correlation, all other α values are low. For the β values, discontinuous urban fabric, green urban and sports areas and barren land have a high correlation. All other class, except agriculture, show a medium correlation to the activity level.

Fitting Parameters for the Multivariate Model As described in the previous section, Khodabandelou, Gauthier, Fiore, El-Yacoubi (2019b) asses a linear relationship between the fitted $\hat{\alpha}$ and $\hat{\beta}$ values and the user activity. This step tries to build a linear regression model to calculate new time-invariant and land use class indifferent parameters. These four new parameters are $\hat{a}_\alpha$, $\hat{b}_\alpha$, $\hat{a}_\beta$ and $\hat{b}_\beta$. The

Class	Pearson α	Pearson β
Continuous urban	-0.625	0.498
Discontinuous urban	-0.107	0.671
Industrial/Commercial	0.142	0.253
Transportation	-0.589	0.364
Agriculture	-0.028	0.026
Green urban	0.305	0.658
Forests/Water	0.23	0.334
Barren land	0.538	0.527

Table 6: Pearson correlation between the average activity level per class, the fitted α values on the logarithmic scale and the fitted β values. Rounded to three digits after zero.

LinearRegression function from the sklearn.linear_model was used, the same that is implemented by the RANSAC regressor for the night population ²⁵. The Ordinary Least squares method was also reused. These parameters are then used in the multivariate model in the next section.

Equations 10 and 11 show the new factors for the multivariate dynamic population prediction model, $\hat{\alpha}$ and $\hat{\beta}$ are the 140 fitted parameters for each land use class from the night population model for the time from 2 am to 7 am in the morning for each weekday. Here, only one of each parameter is fitted, they are not per class or per hour as in the night-population estimation.

$$\hat{\alpha} = \hat{a}_{\alpha} * \lambda(t) + \hat{b}_{\alpha} \quad (10)$$

$$\hat{\beta} = \hat{a}_{\beta} * \lambda(t) + \hat{b}_{\beta} \quad (11)$$

Calculating the Dynamic Population The four new parameters from the previous section allow us to build an estimation model that is able to predict the population in any cell i at any time t on any day. The four fitted parameters $\hat{a}_{\alpha}$, $\hat{b}_{\alpha}$, $\hat{a}_{\beta}$ and $\hat{b}_{\beta}$ are time-invariant and class indifferent, which means they can be pre-calculated and should not change. The model introduced by Khodabandelou, Gauthier, Fiore, El-Yacoubi (2019b) uses multiple different independent variables to predict a single dependent variable, which makes it a multivariate model. σ can be any of three metrics number of calls, Erlang or data usage The intention is the ability to estimate the population of a cell almost instantaneously after receiving the measurement of

²⁵https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html, Accessed 01.02.2023

the number of active users and any of the three possible metrics for σ . The $\lambda_i(t)$ and $\sigma_i(t)$ values can be calculated immediately, and thus the population can be estimated right after. The only values that need to be saved permanently are the four fitted parameters. This approach thus only needs a low amount of data storage and is computationally inexpensive.

Equation 12 shows the estimation model for a cell i at time t . The fitted parameters $\hat{a}_\alpha, \hat{b}_\alpha, \hat{a}_\beta$ and $\hat{b}_\beta$ are used together with the activity at time t in cell i and the chosen metric σ in cell i to calculate the predicted population $\hat{p}_i$ at time t in cell i .

$$\hat{p}_i(t) = e^{\hat{a}_\alpha * \lambda_i(t) + \hat{b}_\alpha} * \sigma_i(t)^{\hat{a}_\beta * \lambda_i(t) + \hat{b}_\beta} \quad (12)$$

To evaluate the dynamic population model, median data from the same four weeks as the night population was chosen to estimate one full week.

5 Results - Validation of Static and Dynamic Population Estimations

This chapter presents the results of applying the models detailed in chapter 4.2 on the selected data set. Beginning with the evaluation of night population estimation, it outlines the used evaluation metrics and n-fold cross-validation. Exemplary night population results are showcased, as well as summed population values and exemplary maps. Then the transferability of fitted models is explored. A new model is trained using Munich data and is evaluated for estimating Munich and other counties population and is then compared to a model trained on the whole study area. The next part is a comparison of the eight different land use classes. Here, the Munich-trained model is again compared against the study-area model. A comparison between the four different used metrics, active users, number of calls, Erlang and data usage and the different hours of the week follows.

The next part is the evaluation of the dynamic population estimation. First, the metric of the Z-score is explained and then the course of a typical day is shown as an example. A comparison of the population estimation with attendance data from mass events follows. The dynamic population estimation uses a model trained on median values from the four week period described in section 4.2.4 to calculate a base-line week. The initial α and β values necessary for the dynamic-population are taken from the nighttime model fitted with nighttime data between 2 am and 7 am. For the comparisons with specific days as in chapters 5.2.3 and 5.2.2 the parameters fitted on the median values is used on the data from the specific date.

5.1 Nighttime Population Estimation

The nighttime population estimation can be evaluated by comparing it to the ground truth census data. But it should be noted that this ground truth data is from 2011 and most likely does not represent the current population accurately. To assess it's quality, the two metrics NRMSE (Normalized Mean Square Root Error) and the determination coefficient R^2 are used combined with a n-fold cross validation.

The methodology described for estimating the night population in 4.2 was first applied to the four week data-set described in chapter 4.2.4, with the data taken from Tuesdays at 10 pm as decided on in chapter 4.2.3. The median values of all Tuesdays at 10 pm from the four week study-period were chosen to eliminate outliers. The resulting model is shown in chapters 5.1.2, 5.1.3, 5.1.4 and 5.1.5. The additional model trained only on data from the city of Munich in chapters 5.1.3 and 5.1.4 is trained with the same data-set, just clipped to the city area.

5.1.1 Evaluation Metrics

To evaluate the model quality, two metrics, the determination coefficient R^2 and the normalized root mean square error (NRMSE) as described in Khodabandelou et al. (2019b) are used.

Determination Coefficient - R^2 The unadjusted determination coefficient R^2 for linear relationships as used in Khodabandelou et al. (2019b) is calculated using the following equation :

$$R^2 = 1 - \frac{\sum_{i=1}^N (p_i - \hat{p}_i)^2}{\sum_{i=1}^N (p_i - \bar{p})^2} \quad (13)$$

Here $\bar{p}$ is the average population density from the ground truth census data per class. p_i is the census population in cell i and $\hat{p}_i$ is the estimated population from the model. The numerator is the model sum of squares, whereas the denominator describes the sum of squared total deviations. N is the count of raster cells. The determination-coefficient ranges between negative infinity and one, where higher values indicate that a higher percentage of the variance in the data is explained by the data. Negative values indicate that the model estimation is worse than taking the average. But a low R^2 could also indicate a non-linear relationship in the data (Schuster, Liesen, 2017:p. 207 f.). The R^2 can be calculated for the whole data-set, or for each of the land use classes separately. It should be noted, that this is the un-adjusted R^2 , which always increases when adding new predictors (Schuster, Liesen, 2017:p. 229 f.).

Normalized Root Mean Square Error - NRMSE The NRMSE measures the difference between the estimated values and the actual ground truth data. It is calculated as shown in equation 14 for each raster cell as also used by Khodabandelou et al. in 2019b. Here N is the total number of cells in the examined values. p_{max} and p_{min} are the maximal and minimal population per class. $\hat{p}_i$ and p_i are the estimated and ground truth population in cell i . The NRMSE can also be calculated for each land use class.

$$NRMSE = \frac{1}{p_{max} - p_{min}} \sqrt{\frac{\sum_{i=1}^N (\hat{p}_i - p_i)^2}{N}} \quad (14)$$

The NRMSE is the normalized form of the Root Mean Square Error (RMSE). The RMSE is well suited to describe normal distributed data. The RMSE is measured in the same units as the data, and thus can not be used to compare different models as easily. That is why the normalized form is used. The NRMSE can lie between

zero and one. Smaller values indicate a smaller unexplained or residual variance and a better fit of the model. The NRMSE is also sensitive to outliers (Hyndman, Koehler, 2006).

K-Fold Cross-Validation To estimate the models ability to predict new data, and not just the data used to fit the model, a cross validation is needed. For a n-fold cross-validation, the data-set is divided into n parts. Each part has roughly the same size. The data is also stratified, that means each of the eight land use classes is represented in the n data sets in about equal proportions. For each round of the cross-validation, one of these parts is selected as the test data-set, whereas the remaining n-1 data-sets are the training data. This means that the model is fitted n times with different selections of data, then the population is estimated using the test data. The evaluation metrics are then calculated using each of the n estimations. The end result is then the average value of the n evaluations (Witten, 2011:p. 152 f.)

5.1.2 Representative Night Population Results

The amount of created data is exhaustive and the study area is large. Therefore it is not possible to go into detail of all aspects of the results. This chapter thus first shows aggregated population numbers for the study area and then illustrates three selected exemplary results.

As an initial quantitative overview, table 7 shows all eight counties in the study area, with their estimated population according to the night population model, their population according to the 2011 census ground truth data and a comparison with the county-level census from 2021. The data is ordered by the 2021 census population. The proportion of the estimation to the census data is also given. For the comparison with the census 2011 data, the estimation is lower for the counties of Munich-City, Munich county and Fürstenfeldbruck, with 67%, 89% and 95% respectively. Freising, Ebersberg, Erding and Starnberg estimated a higher population. The county of Dachau was estimated at almost the same level. The model underestimated the population in the denser, more populated counties and overestimated in the less dense ones. The counties of Fürstenfeldbruck, Dachau and Ebersberg had the most accurate prediction, with a deviation of 5%, 0% and 1%. In total, the model estimated a 16% lower population than the census 2011. The census of 2021 shows a higher population than 2011, resulting in a greater deviation for the estimated population. The estimated population of the dense city of Munich is around 39% smaller than the actual value. Here, all counties except Erding and Starnberg have a lower esti-

mated population than the census population. Freising has the lowest deviation with 2%. In total, the model predicts a 24% smaller population than the census 2021 .

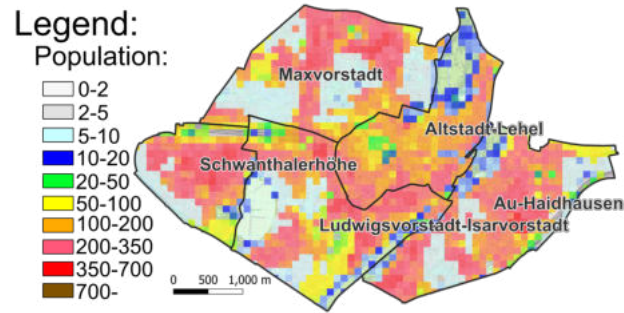
County	Estimation	Census 2011	% of 2011	Census 2021	% of 2021
Munich	912,432	1,348,723	67%	1,487,708	61.33%
County of Munich	284,976	317,832	89%	349,837	81.45%
Fürstenfeldbruck	191,858	201,447	95%	218,579	87.78%
Freising	178,349	163,000	109%	181,144	98.464%
Dachau	138,711	138,652	100%	155,449	89.23%
Ebersberg	129,591	127,598	101%	144,562	89.64%
Erding	149,415	125,990	118%	139,622	107.01%
Starnberg	159,656	127,010	125%	136,747	116.75%
Sum	2,144,988	2,550,252	84.1%	2,813,648	76.23%

Table 7: Estimated population in each county, compared with aggregated population data from the 2011 census and county-level census data from 2021. The proportions of the estimated population to the two censuses are given in percent

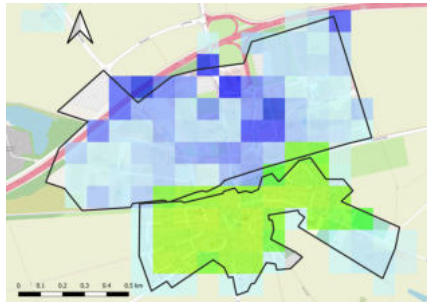
This section shows the predicted population for a few exemplary, smaller areas. One in downtown Munich, one in the city of Freising and one in the village of Parsdorf.

The first selected area is the densely populated city center of Munich. Figure 16a shows the five districts of Ludwigsvorstadt-Isarvorstadt, Maxvorstadt, Schwantalerhöhe, Au-Haidhausen and Altstadt-Lehel of Munich. The color scales from white, grey and light blue for low values, over dark blue, green, yellow and orange for medium population values and dark red and brown for high values. It can be seen that the model estimates a high population of 250 or more for a plurality of 100m by 100m cells. This is significantly higher than the average population density of 48 per cell for all of Munich. The big uncovered area in the west of the Ludwigsvorstadt-Isarvorstadt is the Theresienwiese, a big area without any buildings used for temporary events like the Oktoberfest. Because of the building weighting used in the model, this area has a weight of zero, and thus no predicted population. The same applies to the uncovered area in the north of Altstadt-Lehel, the English garden.

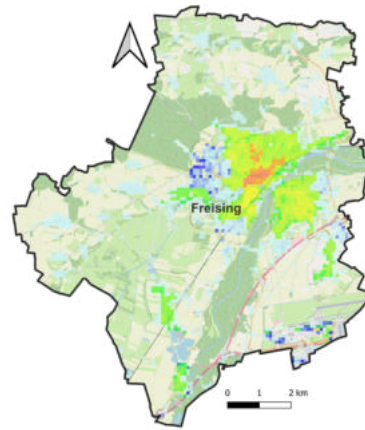
In contrast, the next example, the village of Parsdorf only has a small population, but was chosen because it has a clear division into a residential and an industrial area. Figure 16b shows the village with the population estimation overlaid. The two black shapes indicate the residential area in the south, and the industrial area in the north. The model estimates average values of 20 to 50 residents in the south and one to ten residents in the north. The school in the south-west and the border areas of the residential area also have lower values. It can be assumed that residential areas have a higher population than industrial ones, which the estimation reflects.



(a) Munich city center



(b) Village of Parsdorf. Industrial area in the north, residential in the south



(c) City of Freising

Figure 16: Exemplary night population maps for three different areas

The small city of Freising with its surrounding rural area is shown in figure 16c. This area includes several small hamlets and parts of the Munich airport in the South-East corner. The model estimates a orange, medium population for the city center of Freising, then a surrounding belt of cells with a population around 30 and then an even lower population of one to ten in cells further away. Most of the hamlets also only have estimated populations in the tens. One exception are the villages of Pulling and Acherich in the north, which both have two small centers with population densities of around 30. The airport also has a low estimated population. But most of the area is building-less, and thus has no predicted population.

5.1.3 Comparison between Munich trained Model and Study-Area trained Model

To estimate the effect the large study area has on the model, an additional night-population model was trained only on data from the city of Munich. This model was then used to estimate the population of the city of Munich and its districts. Then, as

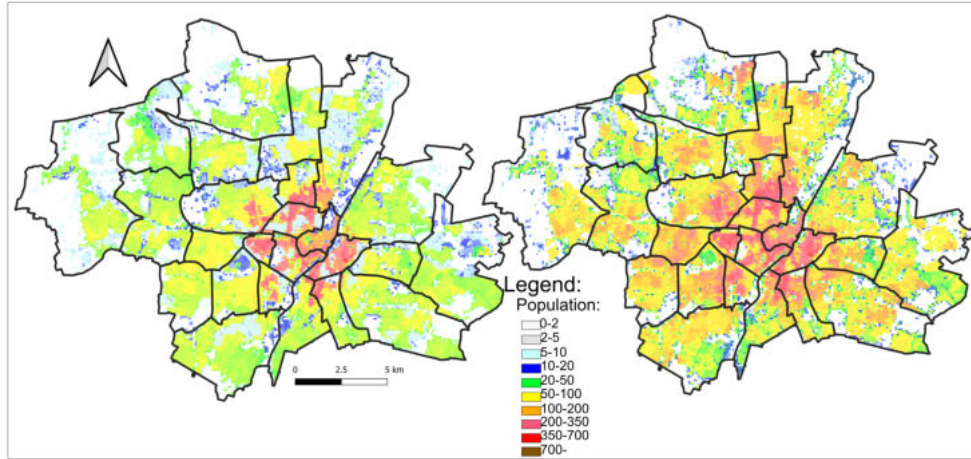


Figure 17: Estimated population for the city of Munich, on the left with a model trained on the whole study area, on the right with a model only trained on the city of Munich

a comparison the same model was also used to estimate the population of the seven other districts in the study area to research the transferability of the models. Then the NRMSE and R^2 values for both estimations were calculated using both model and using a 10-fold cross-validation. This allows for insights into the effect a larger or smaller training data-set has on the model quality.

To compare the two models visually, the map in figure 17 shows the estimated population in Munich from the model trained on the whole study area on the left, and the Munich-only estimation on the right. The districts are marked with a black outline. The Munich-trained model shows a higher population estimate, with a notable concentration of red and orange areas, particularly in the outer districts.

To further evaluate the model quality, the estimated population in each city district was summarized by adding the values of all raster cells in each district all together. These estimations can then be compared with newer 2022 census data from the city of Munich ²⁶. The scatterplot in figure 18 shows the the comparison of the summed estimation and the census data. The Munich-trained points area green, the study-area trained ones are red. As can be noted, Munich-trained model values show a higher, linear correlation with the census data with a few outliers. The red, study-area trained estimation, shows low or no correlation. This indicates that the Munich-trained model has a better fit for Munich. The plot also shows the tendency that for higher census populations, the difference between the two models is also higher, as the two marked districts Ramersdorf-Perlach and Schwabing-Freimann. Conversely, the district with the lowest census population, Altstadt-Lehel, also has

²⁶<https://stadt.muenchen.de/infos/statistik-bevoelkerung.html>, Accessed: 07.05.2023

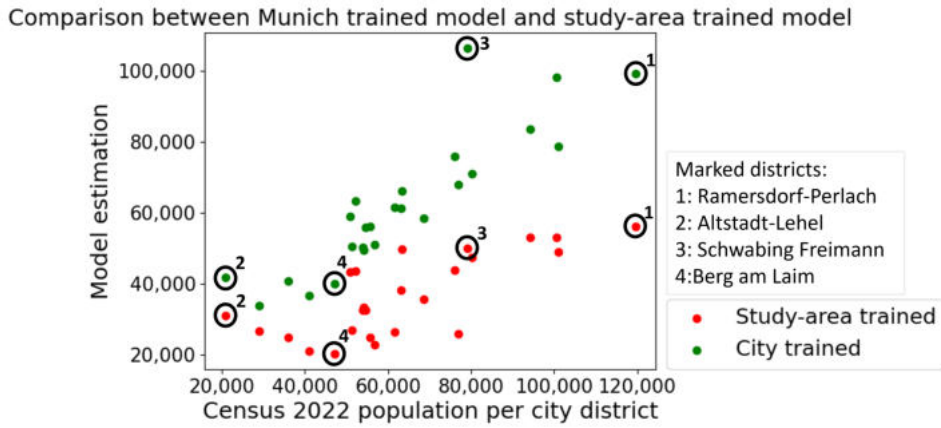


Figure 18: Scatterplot of the estimated population on the x-Axis and the ground truth city district population on the y-Axis. Munich trained results in green, study-area trained in red. Some districts are marked in black.

the lowest difference in model estimation.

To further examine the transferability of a model trained on a single city to other cities, the Munich-trained model was used to estimate the population for the other seven counties in the study area. The comparison between the two models for these counties plus Munich can be seen in table 8. The table also shows the county-level Census population in 2021 and the proportion of the two estimated values to the census data taken from the Bavarian State Office for Statistics²⁷. The comparison shows that for the counties of Fürstentfeldbruck, Ebersberg, Dachau, Erding, Starnberg and Munich, the Munich-trained model performs better. For the county of Munich, and Freising, the study-area trained model performs better. The highest deviations for the Munich-trained model are for the county of Munich and Freising, with 20% and 17% respectively. The study-area model deviates the most for the city of Munich and Fürstentfeldbruck, with 61% and 18% for the city and county of Munich respectively. The highest difference between the two model is for the city of Munich itself, where the study-area trained model misses the 2021 population by 39%, whereas the Munich-trained model deviates only 5%

For further comparison between the two models, the NRMSE and R^2 values for the Munich-trained and Study-area trained models were calculated with the formulas described in section 5.1.1 using a 10-fold cross-validation. In this paper, a value of 10 was used for n , because it is a standard value for limited data sets (Witten, 2011:p. 152 f.). They were calculated for all counties in the study area except Munich, and for the districts of Munich. Table 9 shows the results. For the NRMSE, the Munich-trained model has low values 0.17 for the District estimation, and 0.157 for

²⁷https://www.statistik.bayern.de/produkte/statistik_kommunal/index.html, Accessed: 23.07.2023

County	Munich trained model	Study area trained model	Census 2021	% Munich trained	% Study area trained
Fürstenfeldbruck	223,724	191,858	218,579	102%	88%
County Munich	420,808	284,976	349,837	120%	82%
Ebersberg	137,716	129,591	144,562	95%	90%
Dachau	144,684	138,711	155,449	93%	89%
Freising	211,991	178,349	181,144	117%	99%
Erding	141,476	149,415	139,622	101%	107%
Starnberg	139,743	159,656	136,747	102%	117%
City of Munich	1,558,233	912,432	1,487,708	105%	61%

Table 8: Summed population for eight different counties. First with a model trained only on Munich, then one trained on the whole study area, then Census data from 2021, and then the proportions of the two estimations to the census data

the county-estimation. For the study-area trained model, the NRMSE has a higher value of 0.5 for the districts, and 0.15 for the counties. For the R^2 , the Munich-trained model has medium values of 0.78 and 0.823 for the districts and counties respectively. The R^2 from the study-area trained model in the districts shows a negative value of -0.9, and a value of 0.827 for the cities. A negative R^2 shows that taking the average population values would be a better estimation than the used model.

Area	NRMSE Munich-trained	NRMSE Study-area trained	R^2 Munich trained	R^2 Study-area trained
Munich districts	0.17	0.5	0.78	-0.9
Counties	0.157	0.155	0.823	0.827

Table 9: NRMSE and R^2 values for estimated populations in the different districts of Munich and for the seven counties (except Munich) in the study area. Once for the model trained only with Munich data, and once for the model trained with the whole study area

5.1.4 Comparison between Land use Classes

To compare the eight different land use classes, a n-fold cross validation with the NRMSE and R^2 metrics was carried out. The used data were the Erlang values from Tuesday on 10 pm. The metrics were calculated once for the model trained on the whole study area, and once for the model trained only on the city of Munich. As it can be seen in table 10, there is no barren land class in the city of Munich, which is why these areas were excluded for the population estimation for this model, and

subsequently they also have no NRMSE and R^2 value. All values are rounded to three digits.

For the NRMSE, the values for both the Munich-trained and study-area trained models are almost the same. The lowest values have Agriculture and Forests/Water, with 0.016. Then follow the classes Discontinuous urban, Green urban and Industrial/Commercial with 0.068, 0.07 and 0.07 respectively for the Munich-trained model and 0.06, 0.07 and 0.08 for the Study-area trained model. The barren land class has a value of 0.12 for the study area trained model. Continuous urban has the highest value with 0.2 for both models. This shows an average difference of 20% between ground truth and estimated values. Classes with a higher population density have a higher NRMSE, which indicates a greater difference between ground truth and estimation.

Class	NRMSE Munich- trained	NRMSE Study-area trained	R^2 Munich trained	R^2 Study-area trained
Barren Land	/	0.12	/	0.77
Forests/Water	0.016	0.016	0.61	0.62
Green urban	0.07	0.07	0.84	0.85
Agriculture	0.016	0.016	0.57	0.59
Transportation	0.15	0.1	0.94	0.94
Industrial/Commercial	0.07	0.08	0.86	0.86
Discontinuous urban	0.068	0.06	0.91	0.92
Continuous urban	0.2	0.2	0.99	0.99

Table 10: Results of the 10-fold cross-validation per land use class. NRMSE and R^2 once for the Munich-trained model, and once for the study-area trained model

For the R^2 in table 10 the differences between the two models are small. Agriculture has the lowest value, with 0.57 and 0.59 for the Munich-trained and study-area trained models. Then follows Forest/Water with 0.61 and 0.62, then barren land with 0.77 for the study-area trained model. There is again no R^2 value for the Barren land class of the Munich-trained model. Then follows Green urban with 0.84 and 0.85, Industrial/Commercial with 0.86, Discontinuous urban with 0.91 and 0.92 and transportation with 0.94. The highest R^2 value has the continuous urban class with 0.99, indicating very low amount of unexplained variance. Non of the classes have a low R^2 , which shows that the model is able to explain most of the variance.

5.1.5 Comparison between different Metrics and Hours

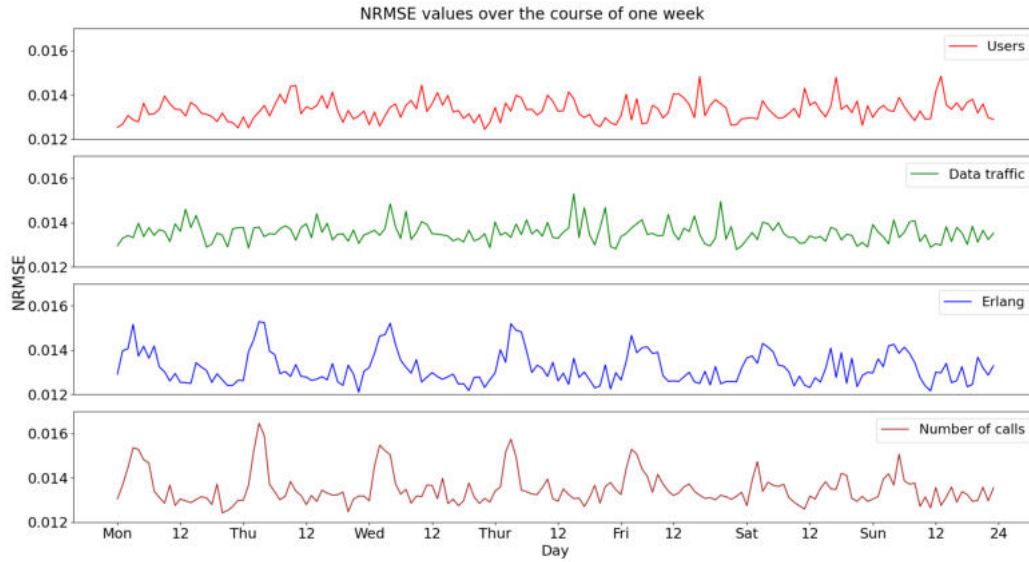
In chapter 4.2.3 it was decided that, according to the pearson correlation coefficient, the Erlang data from Tuesday at 10 pm is the most suited data to estimate the night population. To investigate this decision, a model was fitted with each combination of metric, hour, and day. R^2 and NRMSE values were calculated for each. Again, a 10-fold cross-validation approach was chosen.

Figure 19a shows plots of the NRMSE for the four metrics active users, call amount, Erlang and data usage, over the span of all hours of a week. As a whole, the graph shows a high temporal variation. The Erlang and call amount values reach their minimal points in the evening at around 8-11 pm. Their highest points are in the early morning, between 1 and 6 am. Erlang values are lower than the call amounts. Their NRMSE values are lower on the weekend, than on workdays. Active users peak at 10-11 am in the morning during the workweek, and later at 1-3 pm on the weekend. Their lowest NRMSE values are for 9 pm to 1 am. Data traffic has more homogeneous NRMSE values in the morning between 0 am and 10 am, then a few higher peaks and then again lower values in the night. Weekends have a lower variance. The Erlang values have the lowest NRMSE at Thursday 10 pm.

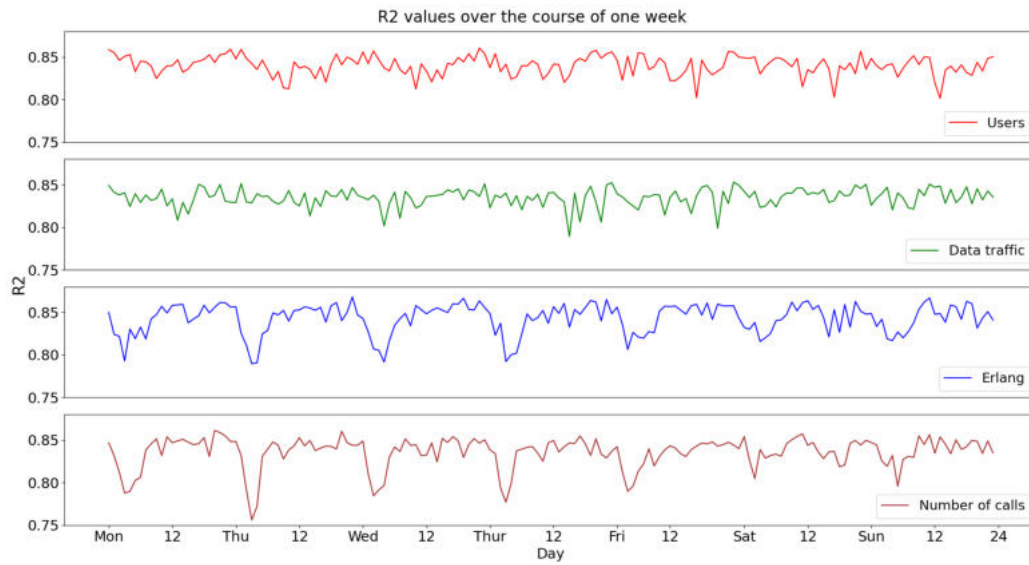
The R^2 results are shown in figure 19b. All R^2 values are over 0.75. Erlang and call amounts have their lowest points in the early morning ours at 3 am. These are lower on the workdays than on the weekends. There is a two peak pattern on the workdays, with one peak around 12 pm, then a drop, and then a second later at 6-8 pm. These two metrics fluctuate more than the other two. The user-traffic values hit their lowest point at midday at around 11 am for Monday, Tuesday and Wednesday and between 12 am and 2 pm on Thursday to Sunday. The peaks are in the evening around 8 pm. On Monday through Thursday, the active user metric has the lowest fluctuation. The data traffic metric peaks in the evening around 6 pm during the workweek and has it's lows in the early morning at around 5 am. On the weekend it has the lowest fluctuation. The highest value of all metrics is achieved on Tuesday at 10 pm for the Erlang values.

5.2 Dynamic Population Estimation

Determining the model quality of the daytime or dynamic population estimation is challenging, because there is no readily available ground truth data to compare the results to. At first, this chapter shows a few exemplary results of the estimation, with a focus on the city of Munich. Then summed population values for the eight counties at different times of day are shown. Subsequently, the population estimations for a few events with known attendance are evaluated. First, a public march



(a) NRMSE values for each metric, for each hour of one week



(b) R^2 values for each metric, for each hour of one week

Figure 19: NRMSE and R^2 values over the course of one week

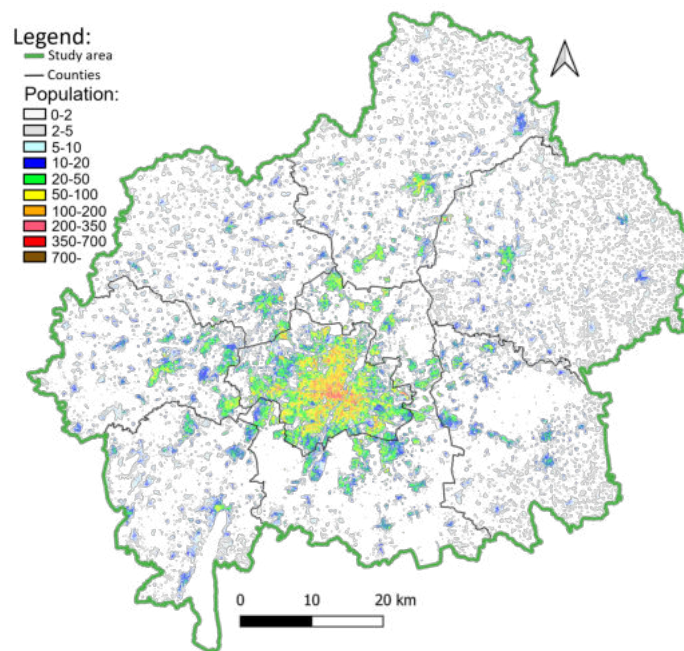


Figure 20: Population estimation on April 19th, at 12 am for the whole study area

with an attendance of between 1,000 and 2,000 persons, then a concert and soccer game with around 70,000 to 80,000 attendees.

5.2.1 Representative Dynamic Population Results

Map 20 shows the whole study area, with the estimated population on April the 19th at 12 am. Blue areas have a low population, red a high one. Areas without overlay have an estimated population of zero. The main city area of Munich as well as many smaller population centers can be seen. The city center of Munich has the highest population density, with many yellow and red areas.

Because it is hard to discern smaller details in a map of the complete study area, the five maps in figure 21 show again April the 19th. Only the Central Munich districts of Ludwigsvorstadt-Isarvorstadt, Maxvorstadt, Schwanthalerhöhe, Au-Haidhausen and Altstadt-Lehel. The five maps show the hours of 5 am, 9 am, 11 am, 6 pm and 11 pm. The red marks show different points of interest. The graph in the bottom right shows the summed estimated population of each city district over the course of the whole day.

All five districts have similar chronological courses, with lower population in the nighttime, then a rising population, starting at 6 am. Their population in the night is

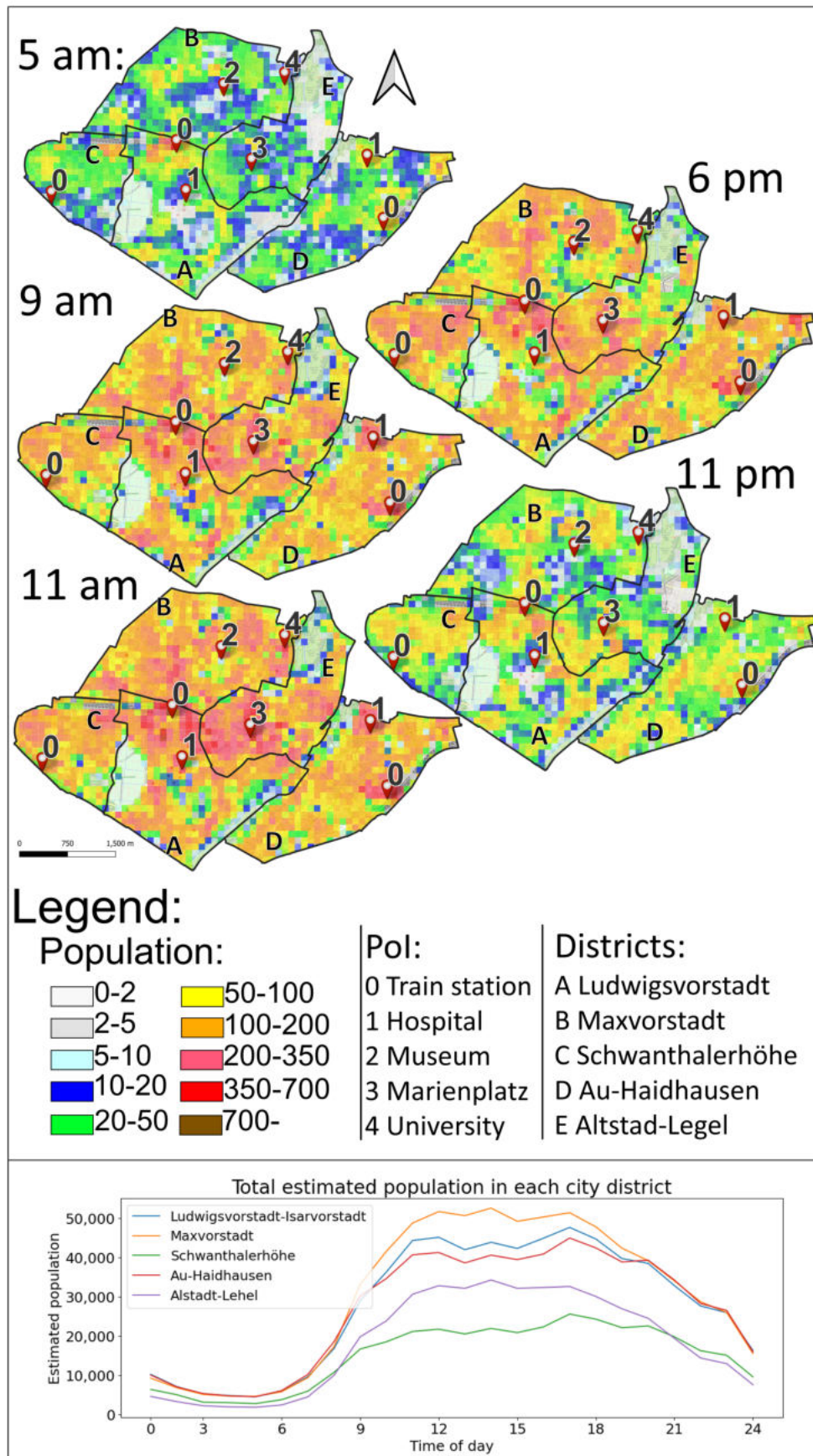


Figure 21: Maps of estimated population in five central Munich city districts at 5 am, 9 am, 11 am, 6 pm and 11 pm. A few Points of Interest (PoI) are highlighted. At the bottom a graph of the population over time per district

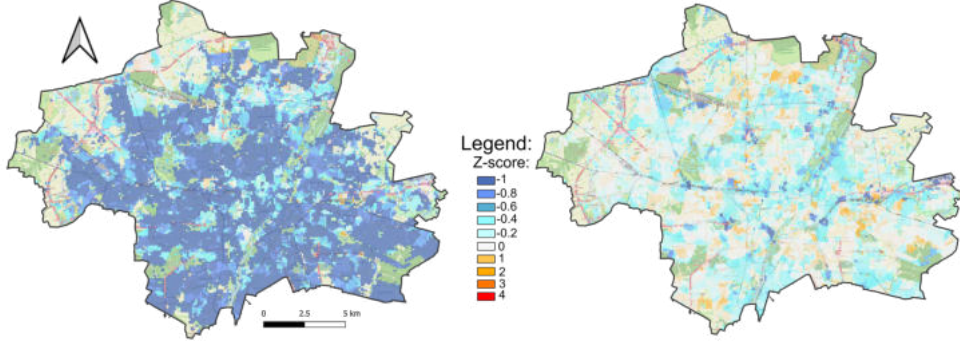


Figure 22: Z-scores for Munich, 6 am on the left and 9 pm on the right

similar. The population in Schwanthalerhöhe stops rising at around 8, and stays at a similar level until lowering again after 6 pm. The other districts have higher peak populations. Maxvorstadt and Altstadt-Lehel have a sharper peak, whereas Au-Haidhausen and Ludwigsvorstadt-Isarvorstadt have a plateau around the midday. These general trends are also visible in the five maps, with a lot of red high-density areas around midday. A few key locations are also visible in the map. For example, the main central station is one of the few areas to have a medium population in the 5am and 11 pm maps.

An additional way of visualizing the dynamic population changes over one day is the z-score as shown in equation 15 and also used by Khodabandelou et al. (2019b). This score denotes the deviation of each cell from the weekly mean value of the population distribution. $\hat{p}_i(t)$ is the estimated population in cell i at time t . μ_i and δ_i are the mean and standard deviation over all values estimated at cell i over the whole estimation time-span of one weeks. A positive z-score indicates that the population in cell i at hour t is higher than on average, a negative z-score indicates the opposite.

$$z_i(t) = \frac{\hat{p}_i(t) - \mu_i}{\delta_i} \quad (15)$$

The map in figure 22 shows the z-score for the city of Munich in the morning at 6am and in the evening at 9 pm on April 19th. Cells with no or only low out and incoming population are white and light blue, cells with high outgoing population are dark blue. Cells with high incoming population are orange and red. As can be seen in the morning, most residential areas have more outgoing traffic, whereas a few transportation hubs and factories show incoming traffic. In the evening, the pattern is reversed. Transportation and industrial areas have more outgoing traffic, some residential areas have more incoming, and a large amount of areas is neutral.

A more quantified approach at evaluating the dynamic population model is aggregating the estimated populations in each of the eight counties in the study area at different times of day. Table 11 shows these values for April the 19th at 5 am, 12 am, 7 pm and 10 pm. As can be seen, the estimated values are significantly lower than the census data for 5am, only reaching between 6% of the census population for Starnberg and 10% for Freising. The estimated population at 12 am and 7 pm is usually between 50% for Erding and 75% for the county of Munich. The 10 pm population is then a again lower, between 36% for Dachau and 55% for Munich. This indicates that the model underestimates the population at all times, but especially in the nighttime.

County	5 am	12 am	7 pm	10 pm	Census 2021
Fürstenfeldbruck	18,707	122,268	129,612	94,566	218,579
County Munich	34,257	264,676	252,982	165,763	349,837
Ebersberg	11,897	73,457	78,829	58,062	144,562
Dachau	12,954	78,027	84,268	57,033	155,449
Freising	19,489	109,413	113,163	84,248	181,144
Erding	12,359	74,260	74,511	52,839	139,622
Starnberg	8,710	68,960	68,028	47,081	136,747
Munich	144,530	1,038,898	1,093,411	795,068	1,487,708

Table 11: Summed population for eight different counties on April 19th, estimated with the dynamic population model. As comparison, the census data from 2021

5.2.2 Comparison with Ground Truth - Public Rally

One area where ground truth data is often available, are public events. One example are two demonstrations in downtown Munich on the April 8th 2023. There was a rally around 12 am at the Marienplatz, with around 600 attendees. At 3 pm, a separate rally at the nearby Odeonsplatz started. Attendees often visited both events ²⁸. The maps in figure 23 show a cutout of downtown Munich. On the left on a Saturday at 12 am, calculated from the median of the four week study period. On the right at 12 am on April 8th. Both rally locations are marked. The black dotted line indicates the route of the march starting at the Marienplatz. The black lines are the outlines of the city districts of Altstadt-Lehel and Ludwigsvorstadt-Isarvorstadt. The march started at 12 am, and lasted about one and a half hours. The background raster indicates the estimated population per raster cell at 12 pm. Blue cells indicate a lower population, reds ones a higher one. As can be seen, there are more red and orange cells around the Marienplatz and the marching route on the day of the rally.

²⁸<https://www.sueddeutsche.de/muenchen/muenchen-ostermarsch-querdenker-demo-1.5787819>, Accessed 20.07.2023

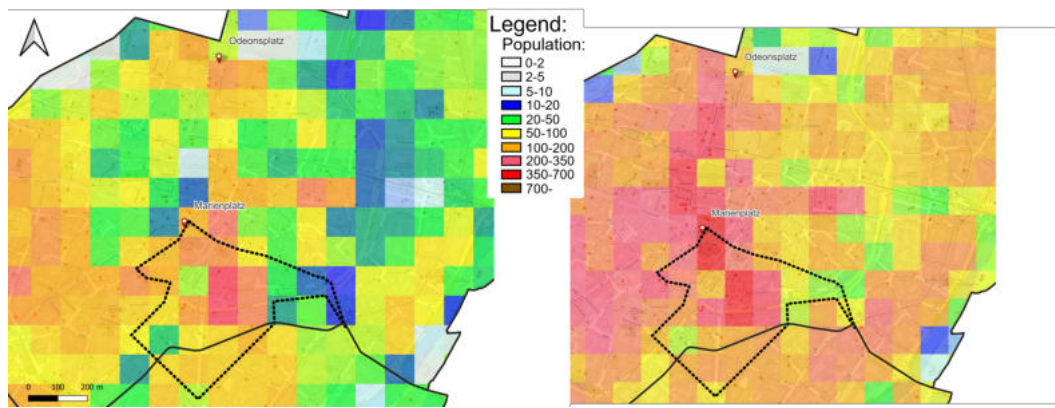


Figure 23: Marienplatz during a demonstration at 12 am on an Saturday without demonstration on the left and during a demonstration on April 8th 2023 at 12 am on the right.

The Odeonsplatz at 3pm has a lower value on April 8th than on the comparison map.

To better quantify the attendance, figure 24 show the estimated population in an rectangular 300m by 300m buffer zone around the Marienplatz on the 8th of April, and the values from the four week median estimation over a whole Saturday. As can be seen, there was a significant higher population on the day of the demonstration at the Marienplatz than on other days. The estimated population rises significantly faster at around 7 am, until it peaks at over 1200 persons at 11 am , the time of the first march. It then reduces to around 1100 attendees at around 6 pm, and then rapidly decreases. At 12 pm the population is the same as on the mean day, around 150. It should be noted that one person could move between different cells within the one-hour sampling period, and thus being counted multiple times in the same hour.

5.2.3 Comparison with Ground Truth - Mass Events

Another area with available ground truth data are mass events like soccer games or concerts. Usually the capacity of the event venues and the attendance is publicly known, and this value can be used to verify the model quality. This chapter examines two mass events, a soccer game at the Allianz Arena and a concert at the Olympiahalle in further detail.

The Allianz arena in the north of Munich has a capacity of around 75,000 attendants²⁹. There is also most likely a undetermined amount of employees active. Thus a

²⁹<https://allianz-arena.com/de/die-arena/fakten-historie/allgemeine-informationen>,
12.07.2023

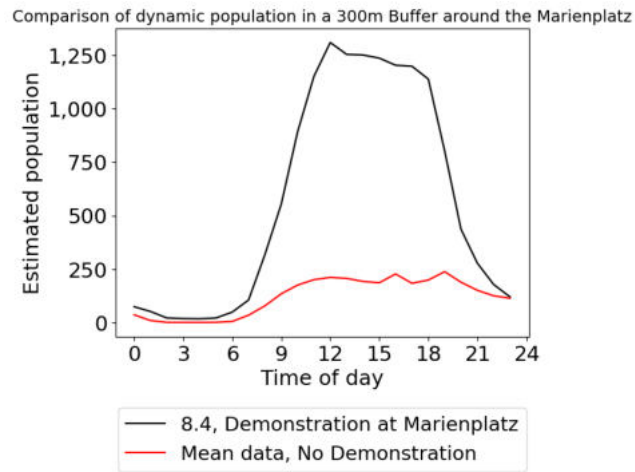


Figure 24: Estimated population at Marienplatz on April 8th in black, and on a median Saturday

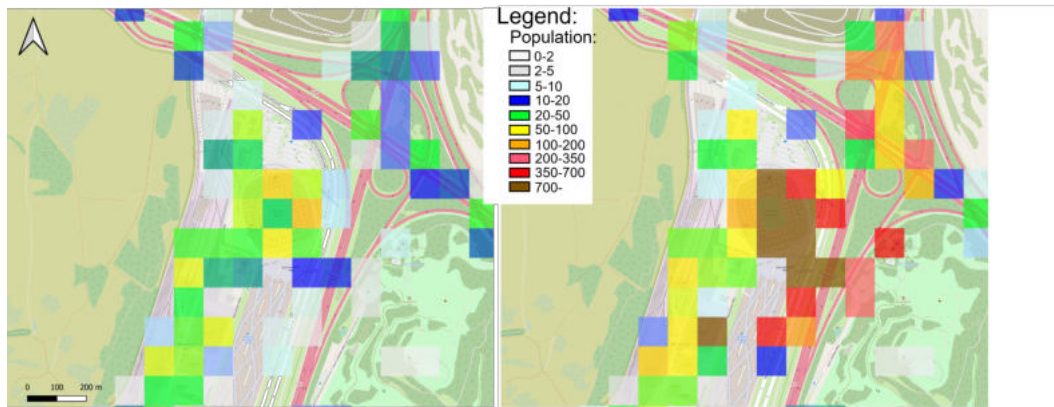


Figure 25: Allianz arena at playtime (right) at 8 pm, and on midday (left) at 11 am

soccer game on April 19th at 8 pm can be used to verify the model. It should also be noted that the Allianz area contains multiple indoor network cells. The map in figure 25 shows the estimated dynamic population values on April 19th at 8 pm on the right, and at 11 am on the left. Again, high estimated populations are shown in red and brown, low ones in blue. As can be seen the estimated population is much higher on the right than on the other day. The population is not only higher in the arena itself, but also in the surrounding area and the train station in the lower left.

The graph in figure 26a shows the estimated population in a rectangular buffer area around the Allianz-Arena. In red, the estimations from the one-week median estimation, in black an estimation using the data from the game on April 19th. Here the estimated population stays the same from 0 am to 4 pm. Then there is a sharp rise in population to around 17,500 at 8 pm. Afterwards there is a similarly sharp decline to the normal population level. Although the estimated population is markedly higher

than on no-game days with an estimated population of around 20,000, the actual ground truth population of the arena should be over 75,000, not including staff.

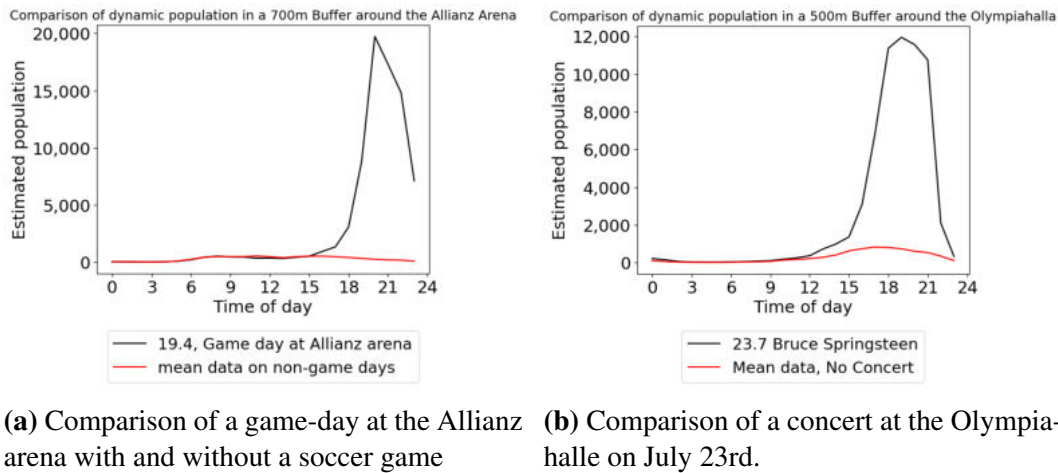


Figure 26: Mass event comparisons

A similar comparison can be made with other mass events. One example is the Olympiastadion in Munich. It has a capacity of 72,000 attendees and can be similarly used to compare the estimated population. The graph in figure 26b shows the population density of a rectangular 500m buffer around the Olympiastadion on Sunday July 23rd in black. On this day a Bruce Springsteen concert took place³⁰. The red line shows the mean population at the Olympiastadion on median Sundays without concerts. The graph is similar to the one from the Allianz Arena. The population on concert day is around 0 until 12 am, then it starts to rise slowly until 3 pm to 1500 persons. Then it raises fast to around 12,000 attendees at 5 pm. It stays around this level until 9 pm, where it starts to lower significantly. At 12 pm, the population is again around zero. The peak is more flattened than the sharp peak of the soccer game. On non-concert days, the population stays at around zero until 12 am, then it raises slowly to a peak of 1000 at 5 pm, and then slowly declines again.

³⁰<https://www.olympiapark.de/de/veranstaltungen/bruce-springsteen-the-e-street-band-n3395>, Accessed: 31.07.2023

6 Discussion - Can Mobile Network Data be used to Estimate the Population?

As mentioned in the beginning of this paper, mobile network data can provide valuable insights into human behavior. Building upon the findings presented in the preceding chapter, the central research question is now addressed: Can the dynamic population be accurately estimated using mobile network data and methods from the field of geoinformatics? Additionally, several sub-questions are addressed: Can the study area be expanded? Can the spatial resolution be enhanced? What is the most suitable network data type? Can raw network performance data be utilized?

As a whole, the used methodology was able to estimate the magnitude and spatial distribution of the static night-population. However, the model predicting the dynamic population only managed to predict relative population levels and changes, and severely underestimated the total population. Many more specific limitations and factors of the two models will be examined upon in the following sections.

6.1 Spatial Distribution of Initial Data

A central point of estimating the human population in higher resolution is the concept of dasymetric mapping. It entails the distribution of data on a coarser spatial resolution to a higher resolution using ancillary data (Järv et al., 2017). This distribution always includes making assumptions about the data, which allow for more precise allocations.

The assumption made in this paper is that people mostly reside in buildings, and thus raster cells containing more building area should be allocated more of the collected network data than cells with less buildings. Broadly speaking, this assumption achieved its desired result and allowed for an estimation with a high spatial resolution, while maintaining accuracy. But it also gave rise to several associated problems. First, the model completely ignores areas without any buildings. This may hold mostly true for the static night population, because a majority of people have their permanent residence inside a building. But this ignores houseless individuals and people living in informal settlements. Decisions made using this model would not take these populations into account and could negatively affect them. For the dynamic population, this approach ignores travellers on streets in non built-up areas, visitors of public parks and rural areas, temporary event locations like festivals and most outside activity. The effect is lessened for inner-city areas, where narrower streets are still covered by raster cells containing adjacent buildings, but it increases while the urban density decreases. A solution could be assigning cells

containing no buildings a small weight depending on the land use class, instead of a weight of zero. Another solution would be using the building weighting for the static population estimation, but a different method for the dynamic one.

A second point of contention is that the model only takes the area of the building footprint into account, not the total usable space including multiple floors. Taking the footprint but not the height into account should work well for areas with homogeneous building heights, but should be less precise in more heterogeneous areas. This effect could be lessened by the different fitted models per land use class, where classes with higher buildings such as continuous urban fabric have different model parameters than areas with more smaller sizes, such as discontinuous urban fabric. The relationship between building volume and population is well established, although there are no studies comparing a footprint based approach with a volume based one (Biljecki et al., 2015). Järv et al. (2017), Greger (2015) and Biljecki et al. (2016) all used different methods incorporating the total floor space of a building in similar weighting function as in this paper. From this follows that a quantitative comparison between these two approaches should be considered in the future.

A further problematic aspect of the building data is the missing information about the function of each building. Not all buildings are residential in nature, there are also commercial, industrial, recreational and other uses. For the night population, only buildings with some degree of residential usage should be considered. For the dynamic population crowd-sourced building function data as in Järv et al. (2017) or business address data as in Greger (2015) could be used as a further weighting function for distributing the network data.

Apart from building data, there is a large amount of other ancillary data that can be used to improve the spatial data distribution. Bakillah et al. (2014) use the density of points of interests as additional data. Sutton, Doll et al. (2006) and Dobson et al. (2000) both use nighttime lights satellite images to enhance their population models. Maantay et al. (2007) uses precise cadastral data. Aasa et al. (2021) use a multi-factored approach, incorporating building data, the street network, land use and information about residential and public buildings into a single model estimating population in Estonia. Järv et al. (2017) use building and land use data, similar to this paper, except they use volumetric building information and they incorporated the land use information into the building data, instead of training a separate model per land use class. The integration of multiple sources of ancillary data appears to be the most promising further research direction. Although the question arises if

the usage of mobile phone data is needed, if the population distribution could be estimated using other sources of data.

A additional point in the distribution of the data is the initial assignment to best server areas. The best server method employed in this paper is most likely comparable to other state-of-the-art methods. The most common way in literature is still the usage of Voronoi or Thiessen polygons, under the assumption that each mobile phone connects with the nearest network site as in Aasa et al. (2021). This assumption ignores the technical realities, where one network site contains antennas with vastly different ranges, and where physical obstacles can block traffic in certain directions and where cell coverage areas can overlap. Ricciato, Coluccia (2023) further emphasize the importance of this initial geo-location of the data. Tennekes, Gootzen, Yvonne A. P. M. (2021) tested more advanced spatial location methods, and came to the conclusion that bayesian propagation models are more accurate than simple tessellations. They also note the possibility of using best server data, but warn that while theses maps are similar to their results, the algorithms used in their creation are usually not accessible, and that they are created with a different goal. This means that the best server method used in this paper is comparable, but does need to be validated in future research.

It can be asserted that the most precise methodology for tessellating space into distinct network areas remains an avenue for further research, but it can be concluded that the simple Voronoi/Thiessen approaches should only be used if no other, more complex, method is usable. Another subject of debate would be the question if the simple assignment of each spatial location to one network location is accurate enough, or if a solution taking overlapping coverage areas and network cell switching into account would produce better results. Building upon this and going back to the research question of the possibility of enhancing the spatial resolution, it follows that this enhancement using a dasymetric approach was achieved in this paper. But the most suitable ancillary data and cell coverage model is still a source of dispute.

6.2 Reflecting Results of the Night Population Estimation

Prior to delving into the results, it is important to clarify the overarching objective of static population estimation, notably its potential role in replacing conventional census methodologies. Traditional census methods are usually based on compulsory, complete interviews on a household-base performed by state officials or mail. The evaluation of the collected data may span several years, and by the time it's completed, the data can already be outdated (Coleman, 2013). For a recent example, the results of the newest census in Germany from 2022 are expected to arrive

in 2024.³¹ While there have been efforts to develop more efficient methods, such as rolling censuses, population registers, or surveys, these alternative approaches often face the same challenge of becoming quickly outdated and inadequately capturing modern, dynamic population trends (Kukutai et al., 2015).

These considerations underscore why it is important to find methods able to replace censuses and provide a dynamic, holistic perspective of populations such as with the mobile network based process in this paper. The results presented in section 5.1 show the applicability of the used linear regression model for this purpose. These results, and the insights about the night population estimation gained from them are discussed in the following sections.

6.2.1 Differences between Models

Table 7 shows the county level aggregated population estimations for the model trained on the whole study area. At a glance, the model was able to differentiate meaningfully between the counties, but it indicates that the model underestimates the population throughout the study area, with a deviation of 24% from the 2021 population and 16% from the 2011 population. The accuracy differs in the different counties. Counties with higher population densities (as shown in table 1) have a lower accuracy, whereas counties with lower densities are more accurate. The difference is the highest for the city of Munich, with an underestimation of almost 40% for the 2021 population.

The model thus does not represent the city population well. One reason could be that the city center of Munich is the most dense area in the whole study area and thus can not be represented well with a model trained on the whole area. This area includes more dense and higher buildings and also contains many areas with heavy pedestrian activity. This is the first indication that a model trained on a large area containing both rural and urban areas is not able to represent areas with different population densities equally well.

The qualitative results for the study area trained model displayed in the maps in figures 16a, 16b and 16c are tenable. With higher populations values in central residential areas in red, with rings of lower density in orange around them. Industrial and commercial areas only show a low population under 10. Areas without buildings show no population, as expected as none of the initial data is spatially distributed to these spaces. From this follows that the model is able to reflect the real-world population successfully. One obvious inconsistency is that the residen-

³¹Source: https://www.zensus2022.de/DE/Aktuelles/Zensus_2022_Ergebnisveroeffentlichung_verschiebt_sich_voraussichtlich_in_den_Maerz_2024.html, Accessed: 23.05.2023

tial population of industrial areas is not zero, as it should be in most cases. An explanation of this phenomenon could be the combined industrial/commercial land use class. Whereas industrial areas usually have no living areas, commercial areas can be of mixed use or interspersed with residential areas. This means a breakdown of this class into two could be sensible in future research. The used land use class information should not be looked at separately from the used building data. As discussed in section 6.1, the building data could be enriched with land use or other ancillary information as in Järv et al. (2017), allowing for a more granular differentiation of different buildings. This could lead to a model which successfully predicts the residential population in industrial areas as zero.

Khodabandelou et al. (2019b) assess that the fitted model values differ between different cities, and notes that while a cross-city usage is possible, the accuracy decreases. Chapter 5.1.3 shows the comparison between a model trained on just Munich, with a model trained on the whole study area. The two maps in figure 17 show tenable results, with the study-area-trained model only estimating low populations in the outer districts of Munich, whereas the Munich-trained model shows more dense populations. The scatter plot in figure 18 confirms this, with the Munich-trained model having a much more obvious linear relationship with the 2022 population, whereas the correlation between the study-area trained model and the census data is low. It is noticeable that the study-area-trained model never made an estimation of over 60,000 residences, which indicates that the model only has a low suitability for high-density cities.

The comparison in table 8 now shows the application of the Munich-trained model on the rest of the study area. Here it can be seen, that for areas apart from Munich, the performance of the two models is more comparable. But a majority of the counties are still better estimated using the Munich-model. This further strengthens the point that the extent of the training data should be carefully selected and the data from the entire area of interest should not be used indiscriminately. The NRMSE and R^2 values from table 9 can be used to verify these results.

For the NRMSE, the values for the Munich-trained model are both low, indicating a good fit, whereas the Study-area trained model only has a low value for the counties, and a medium value for Munich. For the R^2 , the Munich-model has high values of 0.78 and 0.823 for both tests, demonstrating that a high degree of the variability in the data can be explained by the model. The study-model has a unusual R^2 of -0.9 for Munich and 0.827 for the counties. This shows that the Munich-trained model is much better suited to estimate the population in Munich, but only performs slightly worse than the study-area trained model in the other counties. It also shows that the study-trained model should not be used to estimate Munich's population,

firstly because of the low R^2 value which shows that the model is less suited for the prediction than taking the average population value. Secondly, because of the results shown in the previous paragraphs, the study-trained model underestimated Munich's population by around 40%.

It should also be noted that the NRMSE value of the Munich-trained model is higher for the Munich estimation than for the county estimation, and the R^2 is lower for Munich and higher for the counties. Both values show that the Munich-trained model is slightly better at estimating the population in the counties than in Munich itself. This shows that a degree of transferability of models between different spatial areas should be possible, but a model trained on a specific region is better at estimating this region. This is also corroborated in literature, where Ahas et al. (2015) conclude that mobile network activity differs between cities and inside of them. Some cities show earlier activity and some have a larger seasonality. Differences between cities can also point out cultural differences. They also note that the definition of the city area is relevant. Including only high-rise inner city districts leads to different results than also including the suburban areas (Ahas et al., 2015).

Additionally, Douglass et al. (2015) summarize that mobile network activity in cities changes spatially as well as temporally. Population estimation models are only useful in the context of the used network data and should be used together with ancillary data. They recommend to establish a base-line model for each spatial unit of interest, and then re-calibrate this model over time. This approach takes changes in mobile network usage into account (Douglass et al., 2015).

These results generate the question which spatial data sets creates the most generally applicable model, and whether the one model approach is a direction that should be further researched. The temporal variability should also be considered, as there is only a low amount of research into the yearly seasonality of mobile network usage, and its influence on population estimations. As a whole, these results show that the model is able to accurately predict the magnitude of the population, but it does need the right data base to do so.

6.2.2 Contrasting Land Use Classes

The results of the comparison between different land use types are shown in table 10 in chapter 5.1.4. According to the NRMSE values, the model has the highest difference to the ground truth data for the continuous urban and barren land classes. For barren land, the reason could be the small size of the class, which combined with the low mobile network activity can lead to a high susceptibility to outliers. The high value for continuous urban of 0.2 can explain the large difference to ground-truth

data in table 7 in chapter 5.1.2 for the city of Munich. This county mostly consists of dense urban areas and constitutes a majority of this land use class. In general, the model appears to exhibit a tendency of underestimating the population as the density increases. The other land use classes, industrial/commercial, transportation and discontinuous urban have lower NRMSE values around 0.05, indicating a better fit of the model. The two classes with the lowest NRMSE agriculture and forest/water are the two classes with the third and second lowest amount of active users per cell, which could indicate an overfit of the model to the low amount of available data. This information, that areas with lower population are harder to accurately estimate, corroborates with Douglass et al. (2015). This leads to the conclusion that for areas with low population, alternative methodologies should be considered in future research.

The R^2 values show a different picture. The values are quite high, with only "Barren Land", "Forest/Water" and "Agriculture" having values under 0.6. These are the three classes with the lowest amount of active users per cell. But 0.6 still indicates a medium amount of explained variance. The classes "Green urban", "Transportation", "Industrial/Commercial", "Discontinuous urban" and "Continuous urban" all have higher R^2 values with 0.85, 0.92, 0.85, 0.9 and 0.98 respectively. This indicates a high explained variance for all these classes. Especially the "continuous urban fabric" class has an extremely high value, which could also indicate overfitting. Together, the NRMSE and R^2 indicate a model which can explain a majority of the variability in the data, but is not able to make precise predictions and deviates significantly in magnitude from the ground truth. Reasons for this could be outliers distorting the values, heteroscedasticity, which means a non-constant standard error, non-linearity of the underlying data, or measurement error.

A major open question of using predefined land use classes is their comparability to dynamically generated clusters like in Khodabandelou et al. (2019b). They used the median week signature (MWS) method proposed by Furno et al. (2017) to cluster all spatial units with a similar weekly mobile network usage into five distinct land use classes. Similar approaches are presented by Soto, Frías-Martínez (2011), Cici et al. (2015) and Toole et al. (2012). They cluster the data initially into five to eight classes like residential, business, industrial, education or entertainment. Xu et al. (2016) use a similar procedure, but first divide the study area into separate city-blocks using the road network, and then assigning classes per block. They also stress the importance of treating different land uses differently in the population estimation. As they state, the pattern of mobile usage in a residential area is different than the one in a entertainment venue. The classes used in these papers are adapted to only be used in inner city areas, and do not include definitions for rural areas

such as agriculture, forests or lakes. If one wants to allow the population estimation model to work on large areas including rural spaces, these areas need to be included.

It can be thought that the land use classes used in this paper are not dividing the city areas into enough distinct classes, with most cities consisting of discontinuous, continuous and industrial/commercial areas. These do not encompass the diverse urban realms such as entertainment and education and are not distinctive enough. A possible solution for accommodating for both, dense urban areas, and sparse rural areas, could be a two step process. First, identifying urban and rural areas, then clustering the city areas with methods similar to Soto, Frías-Martínez (2011), Cici et al. (2015), Furno et al. (2017) or Xu et al. (2016). The rural areas could then be either divided with another clustering method or predefined classes could be used. The viability of this approach needs to be critically assessed in future research.

6.2.3 Selecting the Best Time Frame and Metric

The results of the comparison of the different hours and metrics that could be used in the night population model shown in figures 19a and 19b in chapter 5.1.5 corroborates the decision made in chapter 4.2.3 to use the Erlang data on Tuesday 10 pm. This selection has the lowest NRMSE and the highest R^2 . On a more general note, the two figures 19a and 19b both show highly regular daily patterns, where only the magnitude of the values changes between days, but not the general course. For the NRMSE, the best results were achieved in the evening times between 6 pm for the calls metric and 1 am for data usage. The least accurate results were those archived a few hours after the lowest point, in the morning hours. The weekends have lower peaks, but similar troughs. One a whole, NRMSE indicates that the model works better for data from the late evening hours. For these late hours the difference between workdays and weekends is minimal. The R^2 in figure 19b leads to similar conclusions. The highest R^2 is achieved in the late evening, and the lowest point is in the early morning. The troughs are also higher on Friday to Sunday than in the rest of the workweek. This means that the model can not produce better results on the weekend, but the minimum performance is better than on workdays. For the differences between metrics, calls and Erlang values both show greater fluctuations than the other two metrics. The data usage metric has a lower magnitude of fluctuation, but a high frequency of peaks followed by troughs. This can lead to the conclusion that the data usage metric is less reliable to estimate population, because it can fluctuate heavily in a short time-frame.

These findings are not corroborated in other works. The German Federal Statistical Office used general network activity data from Sunday between 8 pm and 11 pm as

their basis.³² Khodabandelou et al. (2019b) use weekday data between 4 and 5am. Deville et al. (2014) used a longer time-span, with all hours between 8 pm in the evening and 7 am in the morning being considered.

There is no consensus which combination of metrics and nighttime-hours is the most suited for night-population estimation. It is also doubtful if the different metrics denoting actual mobile network activity can represent the night-population. Most individuals sleep at night, and thus do not create network activity. Although this effect can differ between different metrics. One exception is data traffic, which has a later peak usage than the other metrics, as can be seen in figure 11b in chapter 4.2.3. This could lead to an overestimation of areas with night-life, and an underestimation of other areas. Measurements representing the number of active mobile network devices should, at first glance, provide a more accurate representation of the nighttime population. Because even if an individual is not using the phone at night, it is usually still active and thus counts in the active user metric. An additional contentious issue is the usage of wireless LAN. People at home will often not use the mobile network for data usage and calls, but will connect to their local wireless network. This lowers the amount of traffic in residential areas. Other areas with public wireless LAN like public transport, libraries or entertainment and dining venues could produce similar effects.

Despite this, the results in chapter 4.2.3 do not show that the active users have the highest correlation, even if its pearson correlation coefficient is not significantly lower than the Erlang values. It is also interesting to note the differences between Erlang and the number of calls metric. These two variables are highly correlated, as already shown in figure 10 in chapter 4.2.3. This correlation is logical, but there may be a difference in the call lengths. People may have different proportions of longer and shorter calls at different times of the day. This difference could add an additional dimensionality to the data, allowing for a more accurate model.

A further important point in selecting suitable metrics is future change in mobile network usage. As depicted in figure 5 in chapter 2.4, the utilization of different types of network activity can change significantly over the course of time. Looking ahead, SMS is anticipated to see a further decline in importance. Simultaneously, it is expected that the volume of voice calls and data usage will continue to rise. The adoption of VoIP could also grow, leading to a shift from traditional calls to data usage. Thus the network metric which produces the most precise estimates is subject to change.

³²<https://www.destatis.de/EN/Service/EXSTAT/Datensaetze/mobile-phone-data.html>, Accessed: 20.04.2023

A further disputable aspect is the difference in the definition of the metrics. Whereas the active user and call count metrics can count the same event multiple times by moving between different cells, the data traffic and Erlang measurements can not be counted twice. If a mobile device moves from one cell to another during a call or data usage, the appropriate call minutes and data usage amount are attributed to each network cell. This could indicate that metrics of this kind could lead to more accurate results. It should be questioned if the selection of one, or a few nighttime hours and only one metric to estimate static populations is the most promising way. A different way could be the combination of different data-sources and times. For example, the calls and Erlang metric could be used to calculate a combined median call length metric. Also further study into the time-frame where the highest proportion of the population is at their residence would be useful.

6.3 Reflecting on the Dynamic Population Estimation

The dynamic population estimation has the problem that it is not easily possible to verify the results. It can be suspected that the dynamic model can not produce more accurate results than the static population model, as it is build on top of it. Improving the static model thus should also lead to an improved dynamic model.

All in all, the results of the dynamic population estimation demonstrate the model's capacity to generate reasonable relative population predictions, albeit with some caveats. One specific area of concern pertains to the utilization of building-weighting for spatial data distribution. This distribution may work well for the static model, because it tries to predict residential populations. The dynamic model also tries to predict people on the move and residing at undeveloped locations. Recreational areas like beaches, lakes and parks as well as temporary events like festivals are not accommodated in the model and can't be predicted. An additional problem is the reliance on the activity metric. The mobile network activity of a person can vary wildly between different locations. Whereas the activity may be high during a train commute, it can also be much lower during a concert. This difference in activity necessitates a more granular spatial division into more classes than assumed in this paper. The dynamic separation of the study area into more urban land use classes such as residential, offices, touristic, university or shopping proposed in Khodabandelou et al. (2019b) may be more suited to predicting the dynamic population. The differences in mobile activity patterns could be more pronounced in these classes, than the more traditional land use classes used in this methodology. But this approach poses the question of how to handle less dense and rural areas. There could be differences in the mobile activity in office areas in high-density cities, and offices

in smaller, suburban areas. Additionally comes the challenge of areas with an extremely low amount of active users such as (heavily) forested areas or agriculture. This creates the additional question if a detailed dynamic population estimation on a granular level is even desirable for these areas. For example, estimating a population of five in a forest of 20 km^2 would be better categorized as predicting individual movement, and not populations. One direction here could be the adjustment of spatial granularity to the population density.

The map in figure 20 in chapter 5.2.1 shows tenable results in a spatial sense, with the most dense areas being clearly visible, and lower density areas indicating a lower population. The time comparison in map 21 for the inner city of Munich also shows a reasonable time pattern. The dynamic population is homogeneous in the early morning at 5 am, then the pattern shows more spatial heterogeneity. The only exception is the main train station, which has a higher population at 5 am. At 9 am, few, mostly residential areas in all districts stay at a low population, whereas hotspots with an higher population emerge. First, there are the major train stations of the main train station, the Ostbahnhof and Heimeranplatz. The hospital area in the north-east also has a higher population. At 11 am, the central area between the main train station and Marienplatz is especially busy, with almost none low-population areas. This inner-city area has a large amount of foot traffic and shopping areas. The two university areas in the north and south also now show a higher activity. At 6 pm, the general population is lower than at 11am, but the spatial distribution of the hot spots stays the same. One exception is the district of Schwanthalerhöhe, which has more hotspots in the north at 6 pm than at 11 am. At 11 pm, most cells have a lower population again, a few exceptions are the Marienplatz and the Main train station with slightly higher values. The plot with the population trend over a day, shows that Schwanthalerhöhe has a different pattern than the others. The population raises less over the course of the day, and stays on a similar level from 8 am to 6 pm, without any significant peaks. This could indicate an higher amount of residential areas, with less office spaces, and less foot traffic and commercial activity.

Table 11 in chapter 5.2.1 shows that the dynamic model underestimates populations in all counties. Usually the population at nighttime is underestimated by over 90%, whereas the day time only underestimates between 50% and 25%. This, together with the maps discussed earlier, shows that the model is able to discern spatial and relative differences, but the actual population is not calculated correctly. One reason for the start underestimation at 5 am could be the usage of Erlang measurements. This metric has its lowest pearson correlation and lowest activity in the nighttime between 4 and 5 am, as can be seen in figures 11a and 11b in chapter 4.2.3. From

this can be concluded that while the Erlang metric may be suited for the static or nighttime population estimation, its suitability for dynamic populations in the nighttime is limited. A combination of different usage metrics could lead to more accurate results.

The model used in this paper is adapted from Khodabandelou et al. (2019b). In their work, the dynamic population model showed a similar ability to discern spatial and relative differences. They did not compare the results to census data, but they compared them with the attendance of different soccer games. In their evaluation, their model had an error between -28.4% to 13.2% for 23 different soccer games in three different cities. While these error values are significant, they are lower than the one in this paper. The comparison with mass events in section 5.2.3 shows attendance levels of 20,000 for the soccer game in the Allianz Arena, and 12,000 for a Bruce Springsteen concert. Both venues have a capacity between 70,000 and 80,000, which leads to an underestimation between 75% and 85%.

One aspect for events with high attendance that should not be forgotten is that mobile network cells have a maximum capacity. If the amount of users in a cell exceeds this capacity, new calls and data connections may fail and do not get reported. If all cells in an area are overloaded, it suggests the actual population is higher than the estimate. Thus, cell capacity sets an upper bound on population estimates. Taking an additional data set with the network cell load state into account could alleviate this issue.

But the comparison with an demonstration in the inner city area of Munich again leads to the conclusion that while the model is able to detect outliers and temporary changes in active population, the absolute number is heavily underestimated even in areas with lower dynamic populations. In this specific instance, the observed maximum population of 1300 individuals within a 300m by 300m area centered around Marienplatz appears to be underestimated. Official sources reported an estimated attendance ranging from 1,000 to 2,000 individuals during the march. However, the baseline population figure of 200 appears insufficient for a densely populated inner city area.

The cause of this difference between applications of the model could be the much smaller spatial base-unit, the usage of different data on a raw level or the usage of a predefined land use definition instead of dynamically creating a clustering based on mobile activity patterns. Going back to the research question, the case study shows that dynamic population estimations are possible using methods from the field of geoinformatics, but many questions pertaining temporal and spatial resolution, needed data and auditability are still open.

6.4 Further Limitations

This chapter further focuses on unique limitations pertaining the methodology and used data in this paper.

An initial limitation is the usage of raw performance data. This data, while having the advantage of including zero personally identifiable information, poses several problems. First, because of its low level nature, it is susceptible to measurement errors and missing values due to technical problems. Errors of this kind are often filtered out during further complex processing steps. Often data is supplemented and corrected after a few weeks or months. This did not happen in this paper, because the data was immediately processed after measurement. One aspect of the specific time span of used data in this paper was the low occurrence of these errors. For the dynamic population estimation data from the specific point in time of interest is needed. Missing or faulty data can prevent an estimation completely. On the other hand, one advantage of using this raw data is that the exact definition of the measurements is known, whereas using more processed data can lead to uncertainty about, for example what constitutes a active user. There is also usually a large amount of different raw performance metrics collected, and thus far there is no study comparing them. In summary, raw performance data can be used, but it should be used with caution and a robust error detection methodology.

A second limitation is the missing ground truth data. Measuring the performance of the model on the chosen spatial resolution of 100m by 100m is not possible, because there is no current census data available. An comparison with the 2011 data and using cross-validation can give some insights on model quality. As this data is 12 years old, its ability to determine the quality of the model is limited. This lack of ground truth is especially true for the dynamic population estimation, where no comparison data is available in a general sense. There is some data for attendance of different events like soccer games, festivals or political events. It is not clear if the model performance in estimating these events is reflecting the general model performance, because these events are usually outliers. The mobile network activity at these events could be wildly different than at more typical times, the location of these events is also often in special venues or on large public spaces. This makes determining the model performance in residential areas especially hard. Validating the results using the county and district level census data from 2021 and 2022 can provide some insights into general model performance, but not on the needed spatial level. Even if the aggregated population estimation in any given area is accurate, the spatial distribution inside of the area could still be inaccurate.

A possible further limiting aspect that is usual in any work with a bounded study

area is the possibility of events outside the study area influencing the results. In this case, this problem should be limited to individuals using the mobile network outside of the study area, but inside of a network cell overlapping the study area. Then a part of their measured activity will be counted. But because the edge of the study area is only sparsely populated, the influence on the model should be minimal. Individual travelling in and out of the study area and other migration does not pose a problem to the model. If they are inside the area and are actively using the mobile network, they are accounted for in the model.

The market share is another limitation. The collected mobile network data only includes data from one of the three big network providers in Germany. If the market share is heterogeneous throughout the study area, the model will underestimate populations in areas with lower market share than the average, and overestimate areas with higher market share. This behaviour could be combatted by adding an additional weighting step, where all collected data is weighted by the market share per spatial unit. This does add the additional question of the validity of the market share data. Data of this kind is usually collected using surveys, and can be imprecise. Market share data is also usually only collected from the residents of an area and on a county level, which makes it unsuitable for the dynamic population estimation. Another aspect of using only data from one network provider is that the user base can have a different socio-demographic markup than the general population. DESTATIS reported that by using combined data from Telefónica and Telekom, male customers are over-represented by 13% compared to the 2011 census data, whereas female ones are under-reported by 13%. Different age ranges also have varying representation, with, for example an over-reporting of ages 50-59 by 8%, and an under-reporting of ages 69+ by 13% ³³. To improve this distortion of the population makeup, data from all network providers should be collected, and mobile phone penetration rates of different demographics could be used as ancillary data.

³³<https://www.destatis.de/DE/Service/EXSTAT/Datensaetze/mobilfunkanbieter-strukturvergleich.html>, Accessed: 05.05.2023

7 Conclusion and Outlook on Future research

Within the context of a modern data-rich and dynamic society, the feasibility of estimating dynamic populations from mobile network data presents multiple challenges, but also allows for a deeper understanding of general population trends and behaviors.

In this study, two linear regression models were created to predict the night population as well as the dynamic population using mobile network data. The initial data underwent weighting based on building information and was further divided according to land use. The performance of the various initial data sets active users, voice calls, Erlang and data usage were then compared. The evaluation also probed the usability of different night hours for optimal night population estimation. The validation process for the night population encompassed both the initial spatial resolution, assessed through metrics such as R^2 and NRMSE, and comparisons at the county and city district levels using existing census data. Additionally, the dynamic population estimates were cross-validated against known attendance figures from mass events.

Transitioning from the research methodology to the empirical findings, a first relevant result is that both models, the night and the dynamic population model, were able to estimate the population with a high spatial resolution. One drawback is that the higher resolution lead to less available ground truth data, which made the results hard to verify. This more challenging verification resulted in the data being verified at an aggregated level with newer census data and this comparison lead to the second result that the model tended to under report the magnitude of the population. While the results of the night model were tenable, the results of the dynamic model underestimated the population greatly. One associated insight is that a model trained only on data from the city of Munich was able to asses the population magnitude of Munich more accurately, while still being as good as the standard model at predicting populations in the rest of the study area. This lead to the question of the initial separation of the study area into distinct land use classes based on an available data set. This static separation, instead of a dynamic method based on clusters with similar mobile usage patterns, proved to be usable. Another relevant insight is that the initial spatial distribution of the measured values is of high importance for model performance and at the same time an area with a large amount of different methodologies and approaches. At last, the possibility of using raw network performance data was validated.

These insights lead to the conclusion that this paper effectively addresses the core research question: "Can geoinformatic methods estimate dynamic populations from

mobile network data?" While the answer for this overarching question is yes, the answers for the sub-questions are more differentiated. The possibility of applying an existing method onto a higher spatial resolution and larger study area is indeed present. But the higher spatial resolution leads to a harder verification process, whereas the larger study area points out the question of transferability between models trained on different data sets. The question about the suitability of raw network data can also be answered with yes, with the caveat that this data can contain more errors and missing data than more processed, high level, data. On the other hand, this data allows for privacy-compliance, which may be an even more important factor in the future. The question of the best suited network data can only be answered inconclusively. While the Erlang data produced the best results, this insight is not reproduced in the relevant literature. This is interrelated with the fact that other papers usually use more processed data and there are no current research efforts into comparing these different metrics. The paper also successfully follows up on the work of Khodabandelou et al.(2019b), showing that their methodology is able to be adapted to other data sets.

There are still a plethora of issues that future research might address. While the used methods demonstrate the potential to estimate the dynamic population from mobile network data, the absence of a standardized model and the ongoing debate within the scientific community regarding the optimal data selection, pre-processing techniques, spatial weighting strategies, and prediction models underscore the need for continued research and refinement in this domain. As indicated by the results, all these factors have a substantial impact on the model's quality. The methodology outlined in this paper can be roughly divided into five potential directions for future research.

At first, it would be a great advancement if future research could decide which kind data is the most suited for population estimations. Here it should be considered that the network usage patterns will change over time, leading to shifts in the importance between voice calls, data usage and SMS. With modern digital mobile network infrastructure it is easier than ever to measure a multitude of different metrics. Deciding on the most effective one, and weighing aspects such as privacy compliance will be important.

A second important direction is the initial dasymetric data distribution. There are a multitude of different possible approaches with different initial ancillary data requirements. Works comparing different approaches could help enhancing the spatial resolution significantly and would be of great importance for more detailed estimations.

An additional critical field is the division of the studied area into land use classes or clusters. An initial research question could be if a static division as used in this paper is sufficient, or if a dynamic clustering method should be used. Further research queries could then be the optimal number of distinct classes, differences between rural and urban areas and differences between cities.

The next question concerns the night population model. Here there is already some consensus on using the same linear regression method as used in this paper for the estimation, but there are still some open future research avenues: The transferability of models between different cities and, the differences between rural and urban agglomerations, if the model needs to be retrained in certain time intervals to maintain correctness, the most accurate selection of initial data and the influence of seasonality on model performance. Answering these questions could greatly improve the practicality of using these model in a real-world application.

The last open research question is the dynamic population estimation. Here, a multitude of potential research directions remains open, with no discernible optimal path. The existing regression techniques akin to this study might undergo refinements toward standardization, or emerging machine learning methods employing neural networks could potentially outperform them.

Answering these research questions and being able to dynamically estimate the population on a granular spatial level will allow for the possibility of a wide range of new applications and insights. One application is the implementation of energy-saving measures by dynamically deactivating network cells. Cells who, in a certain time frame, only provide coverage to no or only a low population could be deactivated during this period. This could lead to significant energy savings without any noticeable impact on service quality. The estimations could also revolutionize city planning and disaster relief. Knowledge about the temporal and spatial distribution of city inhabitants could guide placement of infrastructure, public services, and commercial zones. This dynamic estimation could uncover unknown population hotspots, which would benefit the planning of disaster relief and emergency response.

The data driven estimation approach could also be integrated into new smart city and digital twin projects, allowing for an almost real time view of population dynamics and trends. This would also further allow for new insights into societies reaction to events like pandemics, political turmoil and other unforeseen situations. All in all it can be said that it will be exciting to see where the whole field leads to and if data driven approaches as in this paper will deepen our understanding of the world.

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Declaration of Originality

I confirm that the submitted thesis is original work and was written by me without further assistance. The thesis was not examined before, nor has it been published. Appropriate credit has been given where reference has been made to the work of others.

Eidesstattliche Erklärung

Ich versichere, dass ich die vorliegende Arbeit ohne fremde Hilfe und ohne Benutzung anderer als der angegebenen Quellen angefertigt habe, und dass die Arbeit in gleicher oder ähnlicher Form noch keiner anderen Prüfungsbehörde vorgelegen hat. Alle Ausführungen der Arbeit, die wörtlich oder sinngemäß übernommen wurden, sind als solche gekennzeichnet.

Daniel Vester

Augsburg, November 18, 2023

Augsburg, den 18. November 2023